

Nyquist Rate DACs

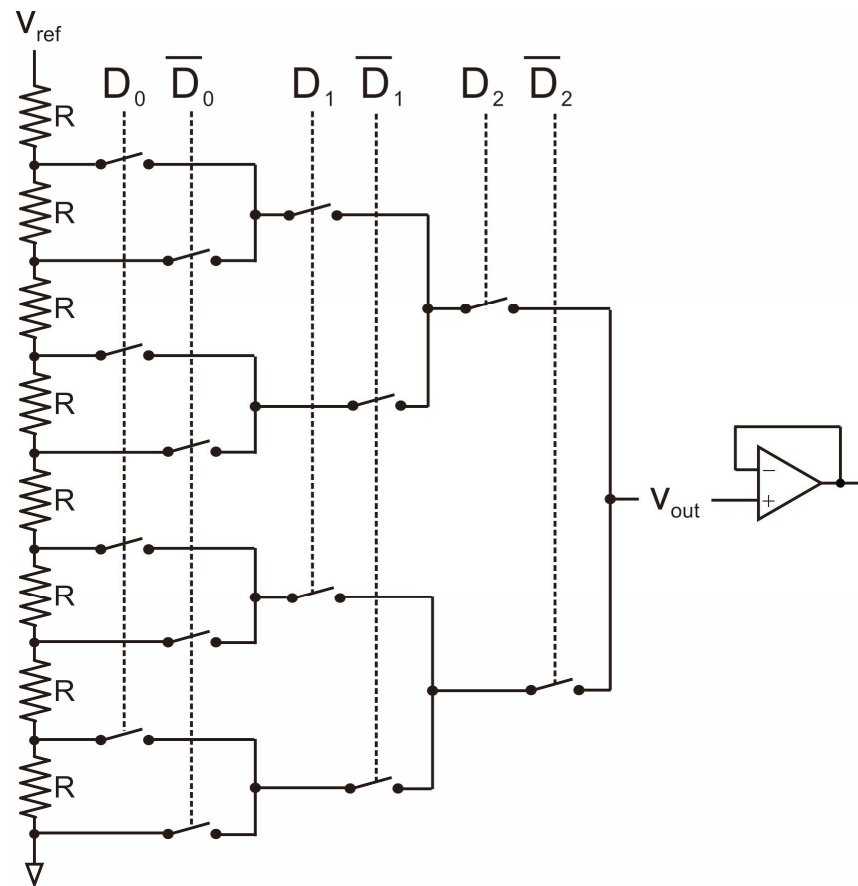
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Overview

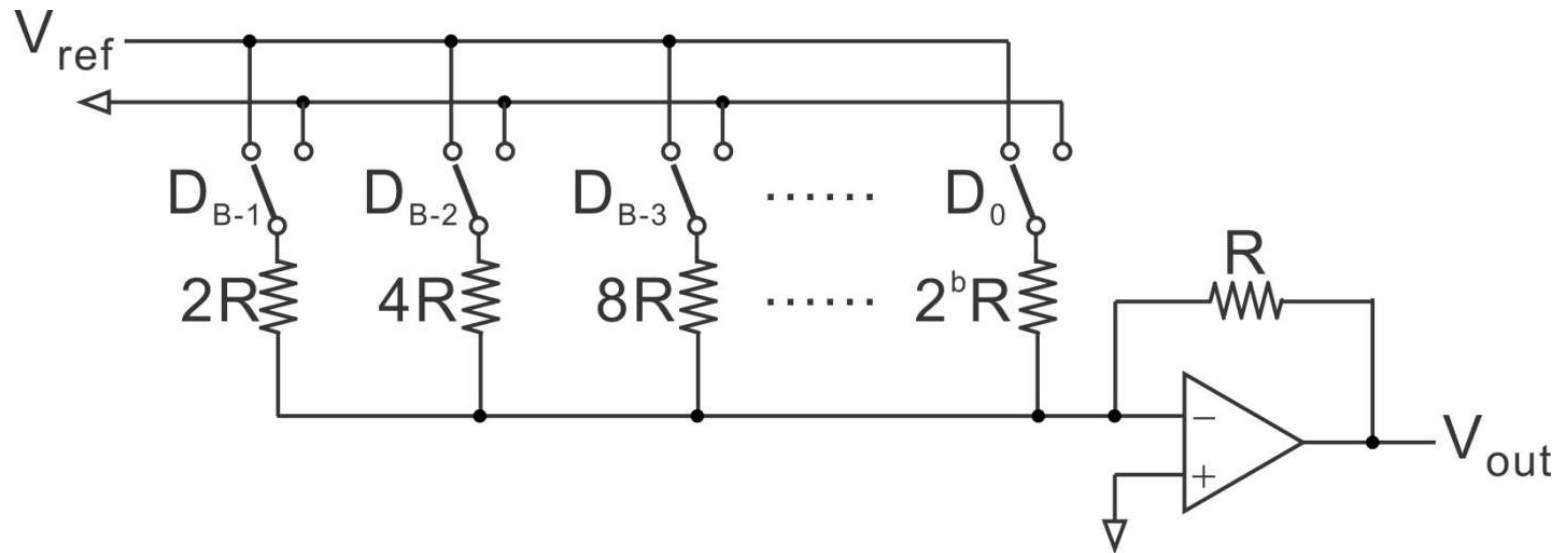
- Topologies
 - Resistor string DAC
 - Voltage switching DAC
 - Current steering DAC
 - Switched capacitor DAC
 - R-2R DAC
- Current steering DAC architectures
- Static errors
- Dynamic errors
- Example implementations

Resistor-String Voltage DAC

- Digitally controlled potentiometer
- Advantages
 - Simple ☺
 - Monotonic ☺
- Unsuitable for high-speed ☹
 - Tree-type of decoder has a large resistance
 - Large and unequal RC time constant at the output
 - Add an opamp will help
 - High power & low-speed
- Unsuitable for high-resolution ☹
 - Too many resistors & switches

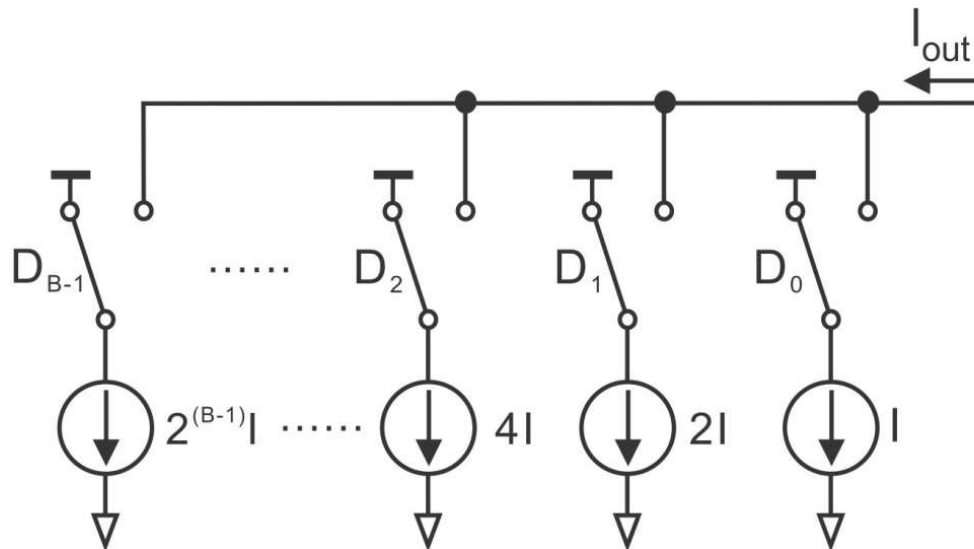


Voltage Switching DAC



- Use analog adders
- Nonlinearity due to nonzero switch turn-on resistance

Current Steering DAC

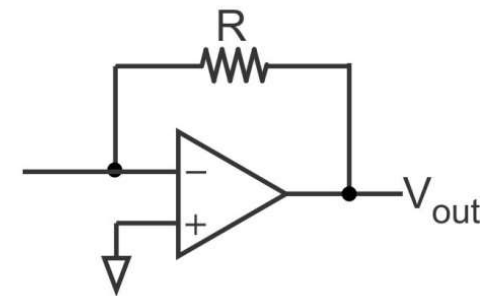


For voltage output

Scheme 1

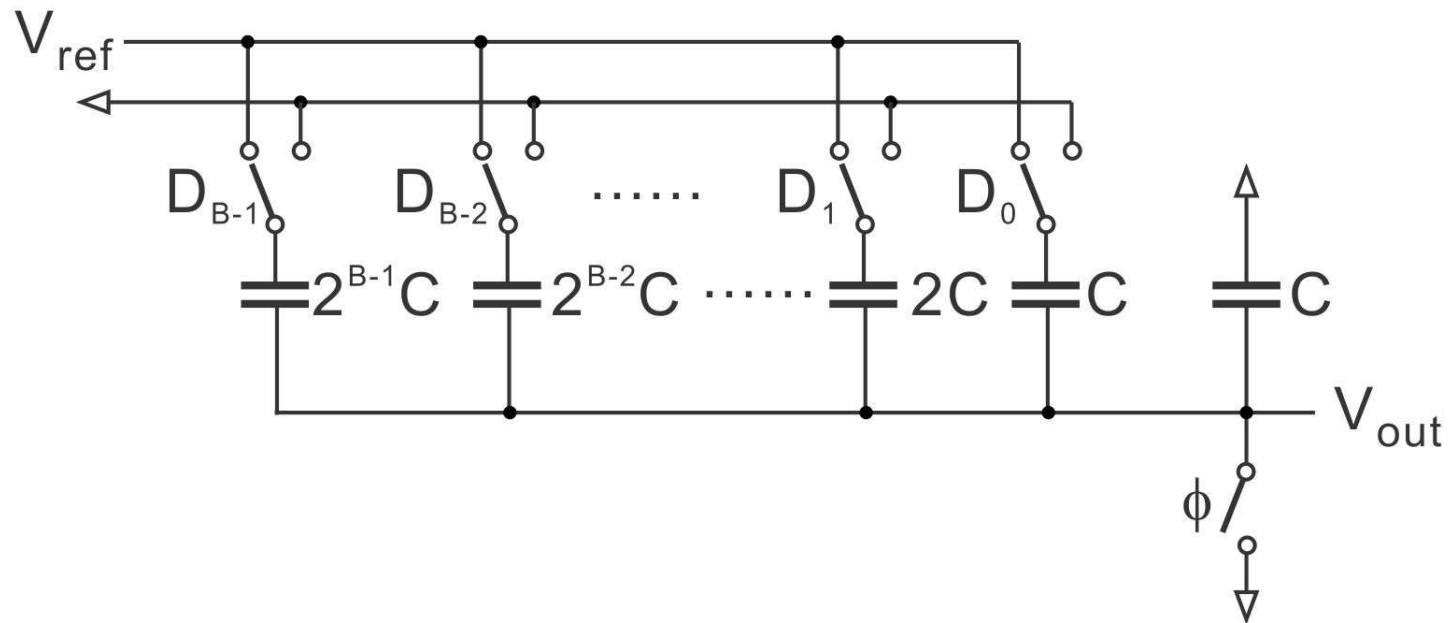


Scheme 2

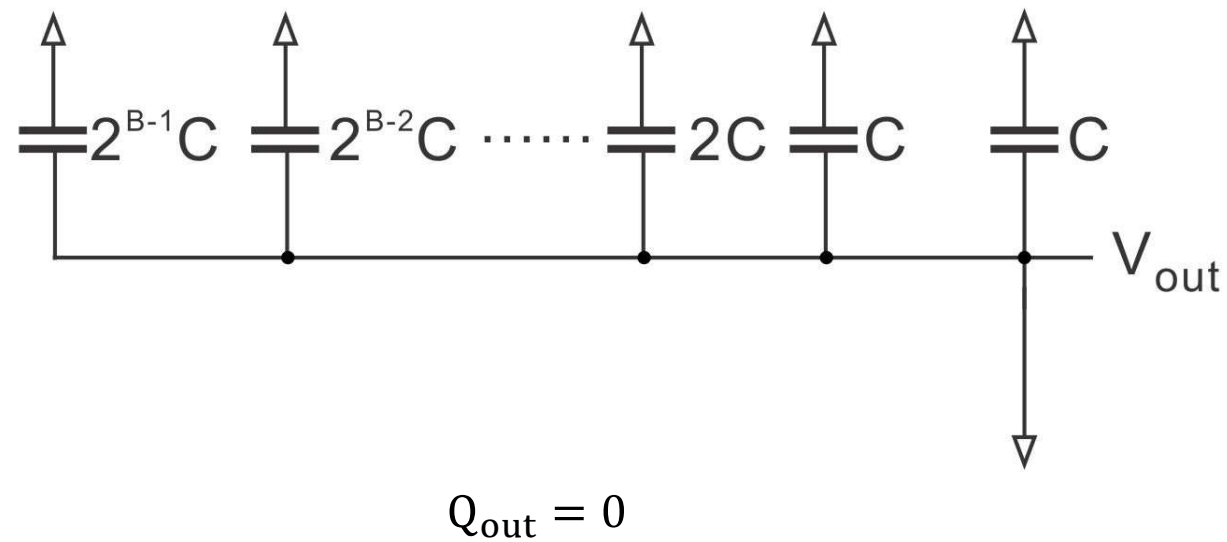


- MOS switches are much better at switching current than voltage!
- Can build very good current sources with high impedances
- Suitable for high speed ☺
 - Avoid the use of opamps
 - Small RC time constants at the output

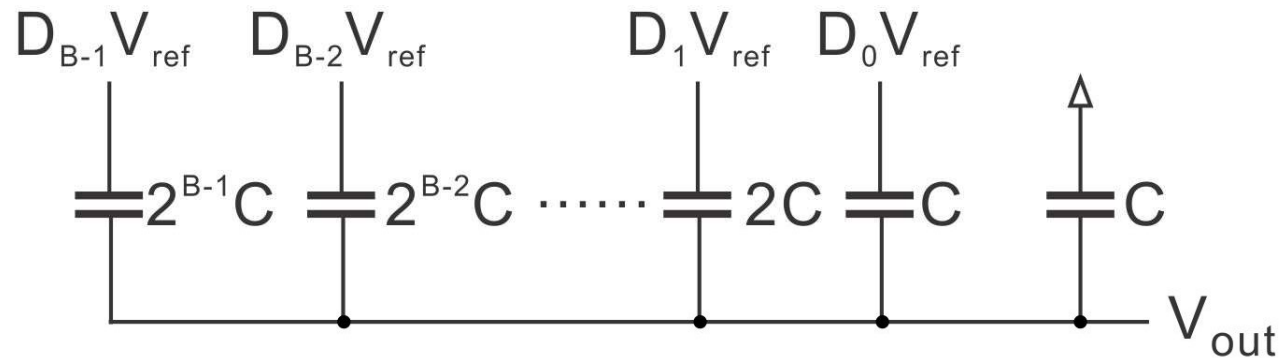
Switched Capacitor DAC



Initialization Phase ($\phi = V_{dd}$)



Evaluation Phase ($\phi=0$)

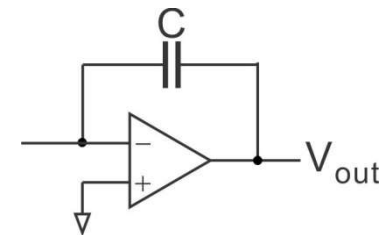


- Due to charge conservation:

$$Q_{out} = CV_{out} + \sum_{i=0}^{B-1} (V_{out} - D_i V_{ref}) 2^i C = 0$$

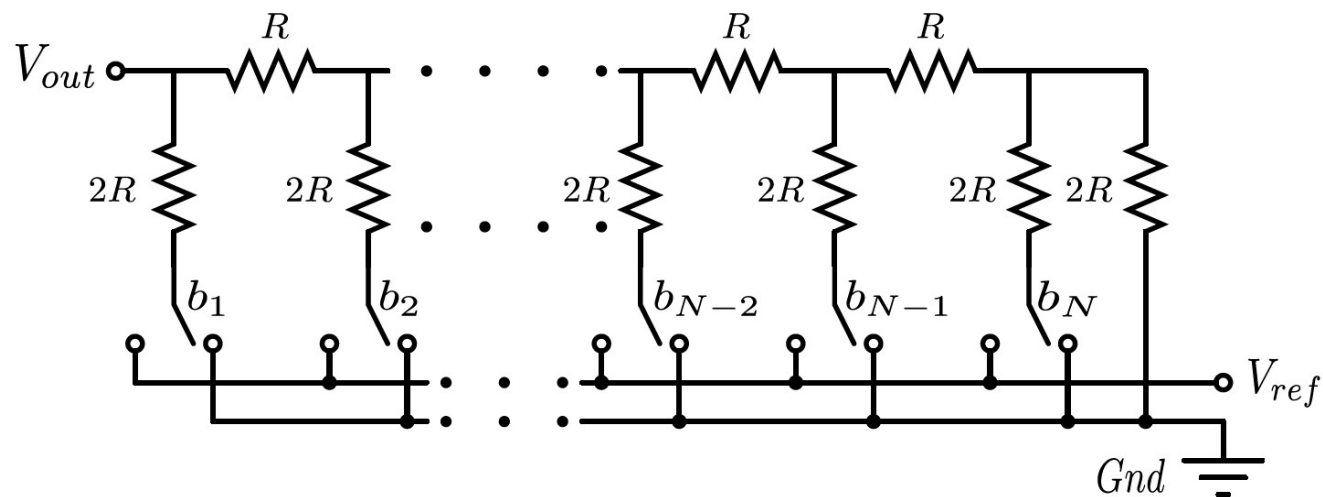
$$V_{out} = V_{ref} \sum_{i=0}^{B-1} 2^{i-B} D_i$$

- Nonlinear cap at V_{out} is a big problem
 - Use an opamp with a feedback cap



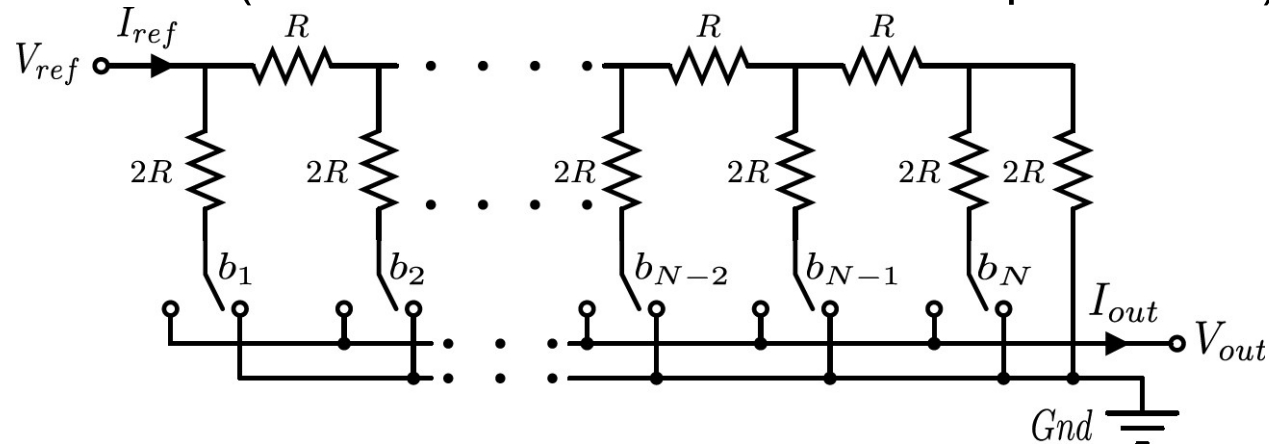
R-2R DAC Concept (voltage mode)

- Aforementioned techniques all require large device ratios
 - Assuming the size of the smallest device is fixed by matching requirement, a large device ratio leads to a large area
- Is it possible to get exponentially weighted voltage/currents by using similar sized devices?
 - R-2R ladder

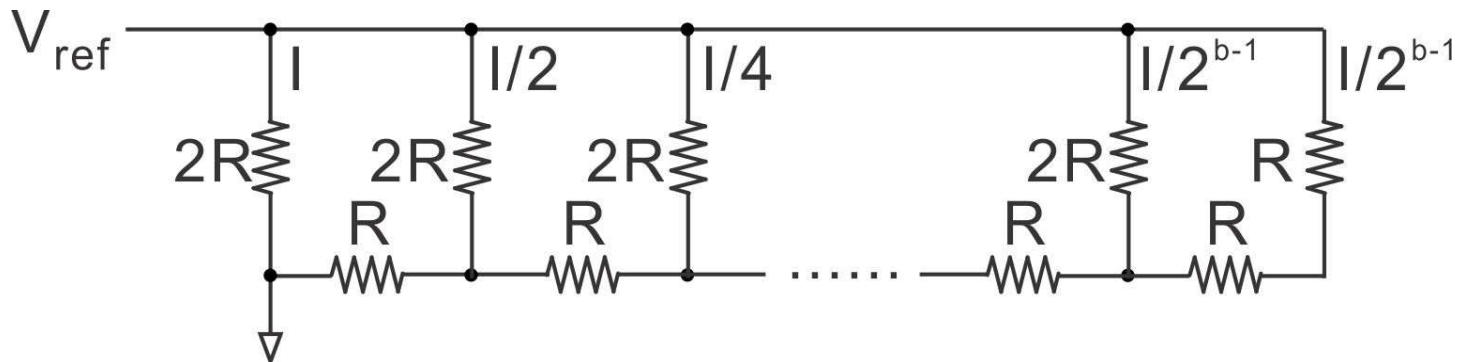


R-2R DAC Concept (current mode)

- Current mode (need a FB-OTA to buffer the output current)

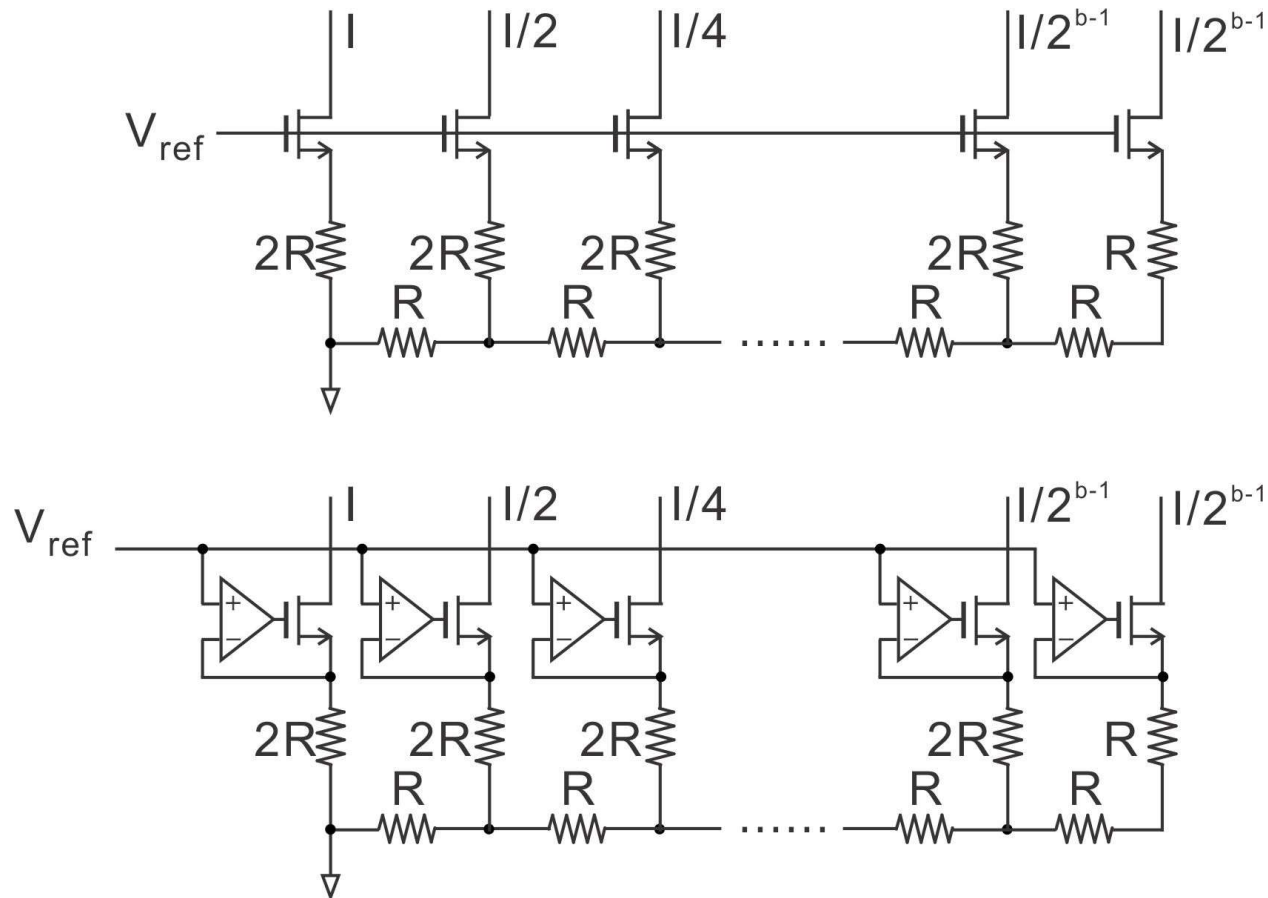


- Output impedance is code dependent



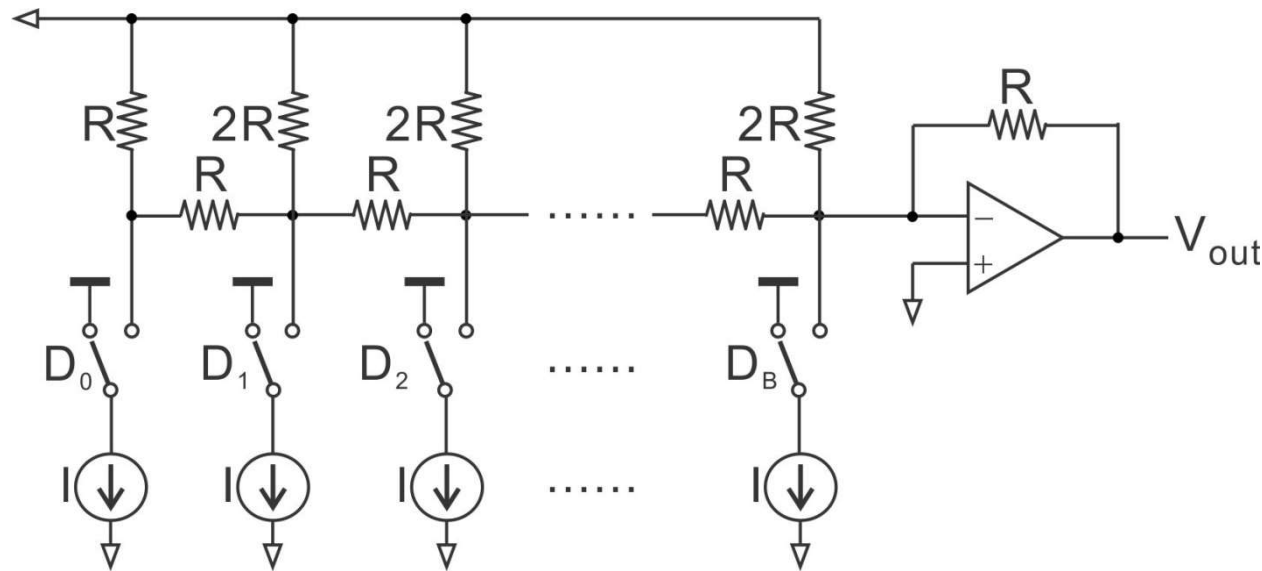
- How to pull the current out?

R-2R DAC Implementation



- OTAs can be used to remove V_{GS} variations
 - Extra power, area, and slow

Drain R-2R Ladder

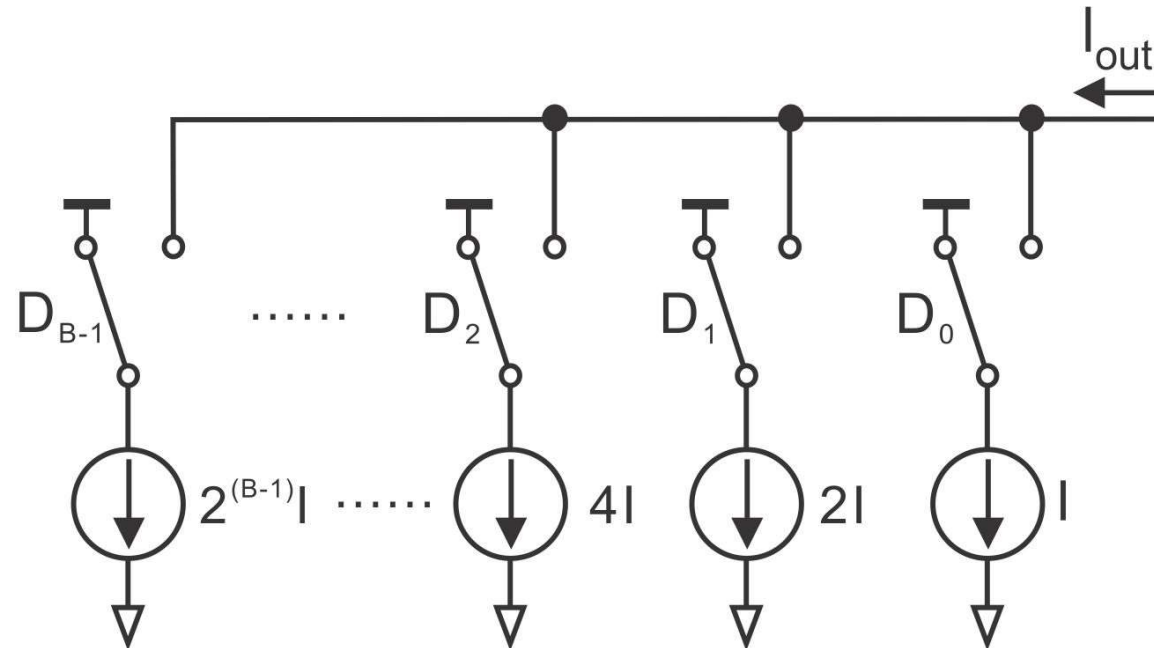


- This topology ensures the same current
- Low output impedance of R-2R ladder $\rightarrow$ necessitate opamp $\rightarrow$ unsuitable for high speed
- Nowadays, most high-speed DACs are current-steering DACs
 - We will focus on current-steering DACs from now on

Overview

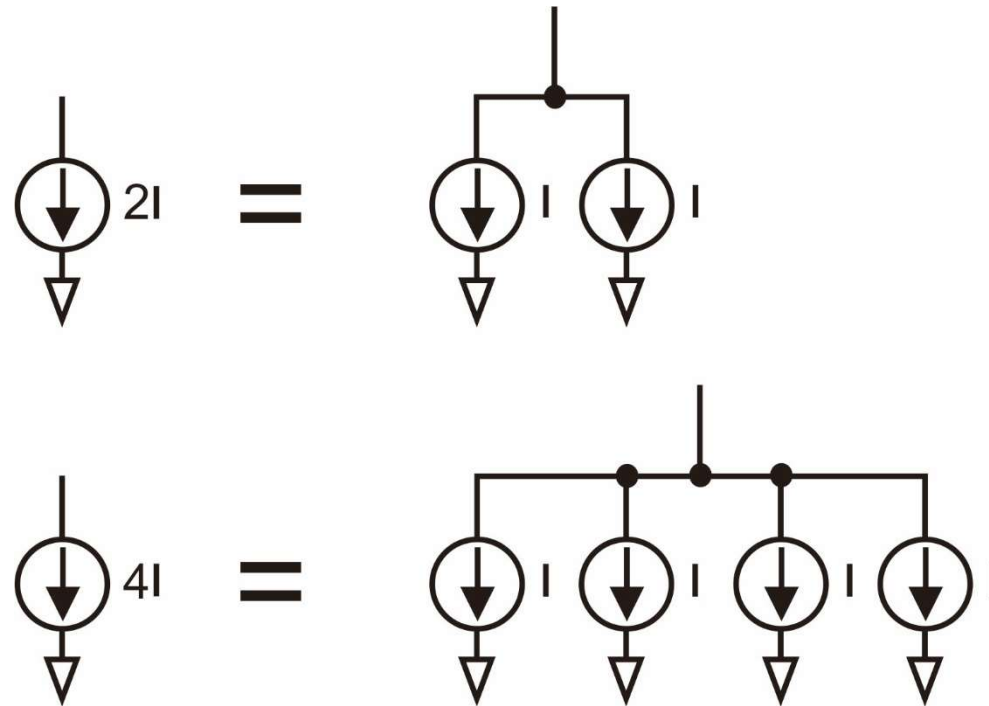
- Topologies
- Current steering DAC architectures
 - Binary weighted DAC
 - Thermometer DAC
 - Segmented DAC
- Static errors
- Dynamic errors
- Example implementations

Binary Weighted Current Steering DAC



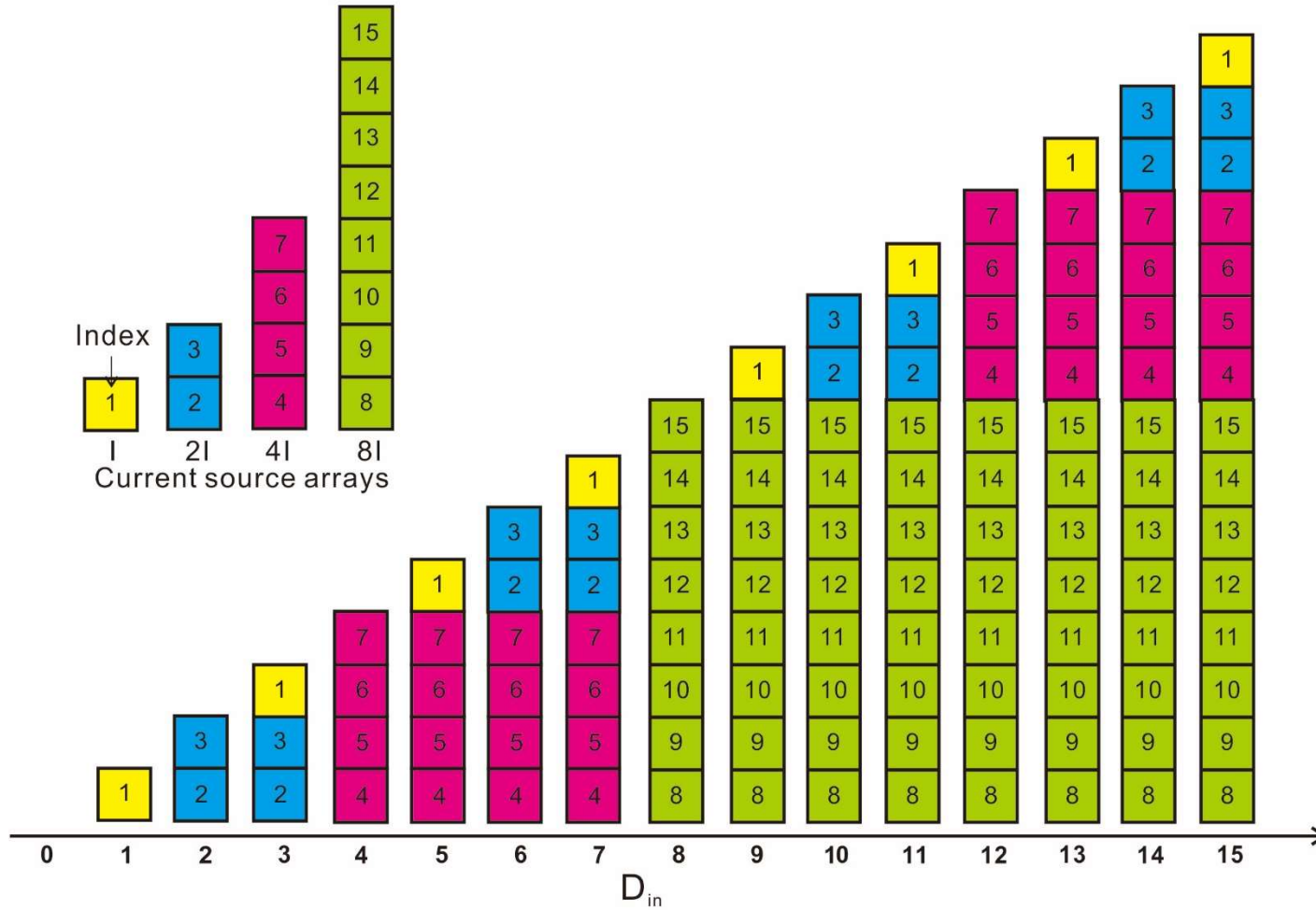
- Simple structure 😊
- No encoder needed 😊
- How to match the current sources?

How to Achieve Good Matching



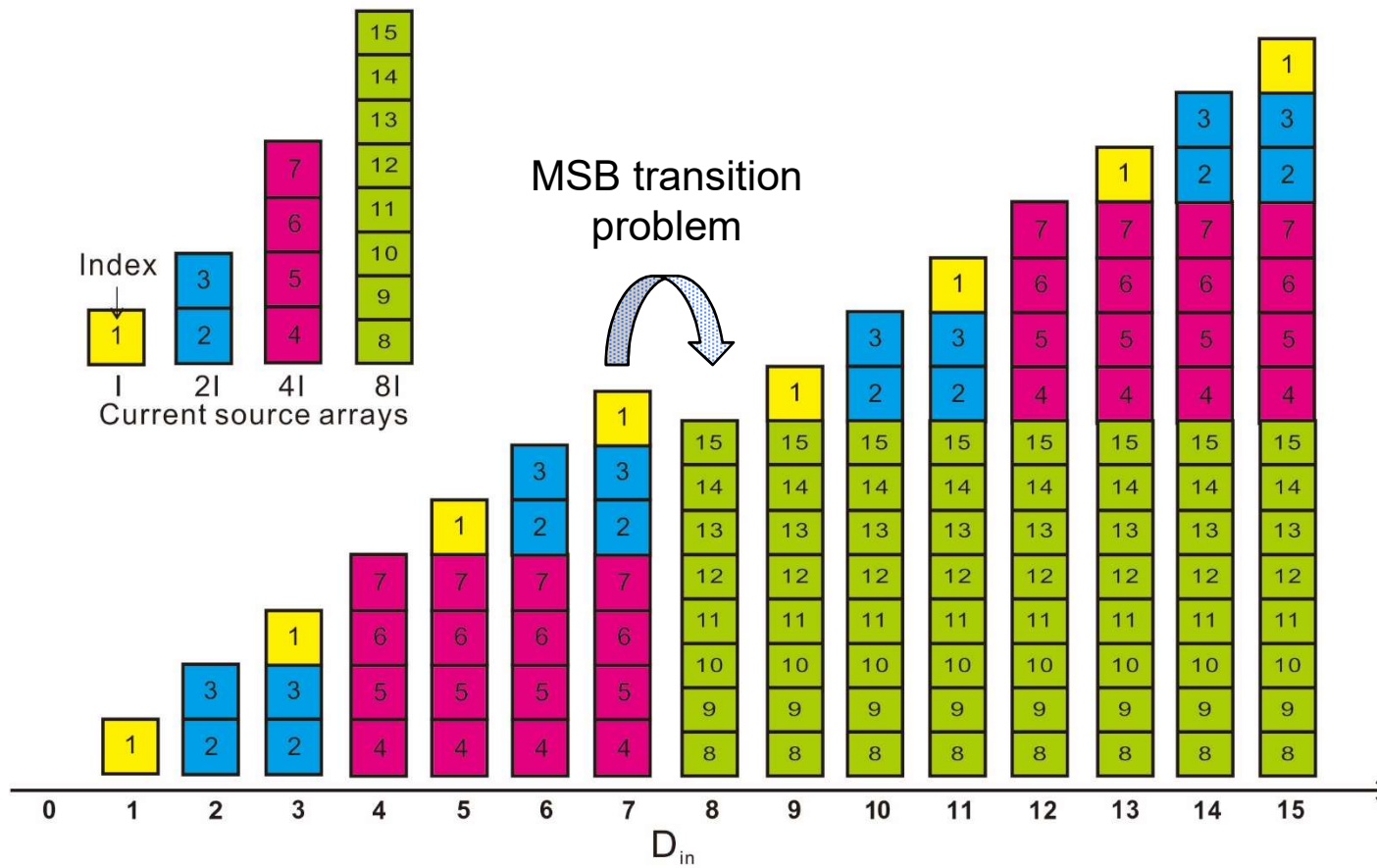
- All current sources are made by identical unit current sources put in parallel.

Binary Weighted DAC Principle

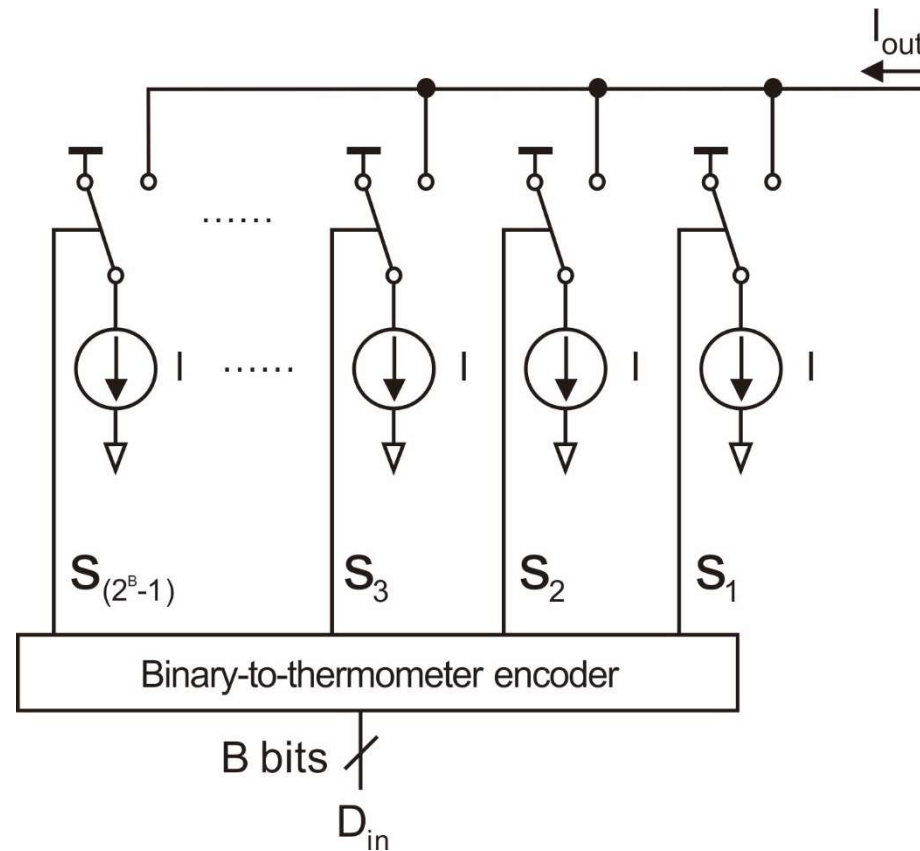


MSB Transition Problem

Cannot ensure monotonicity ☹

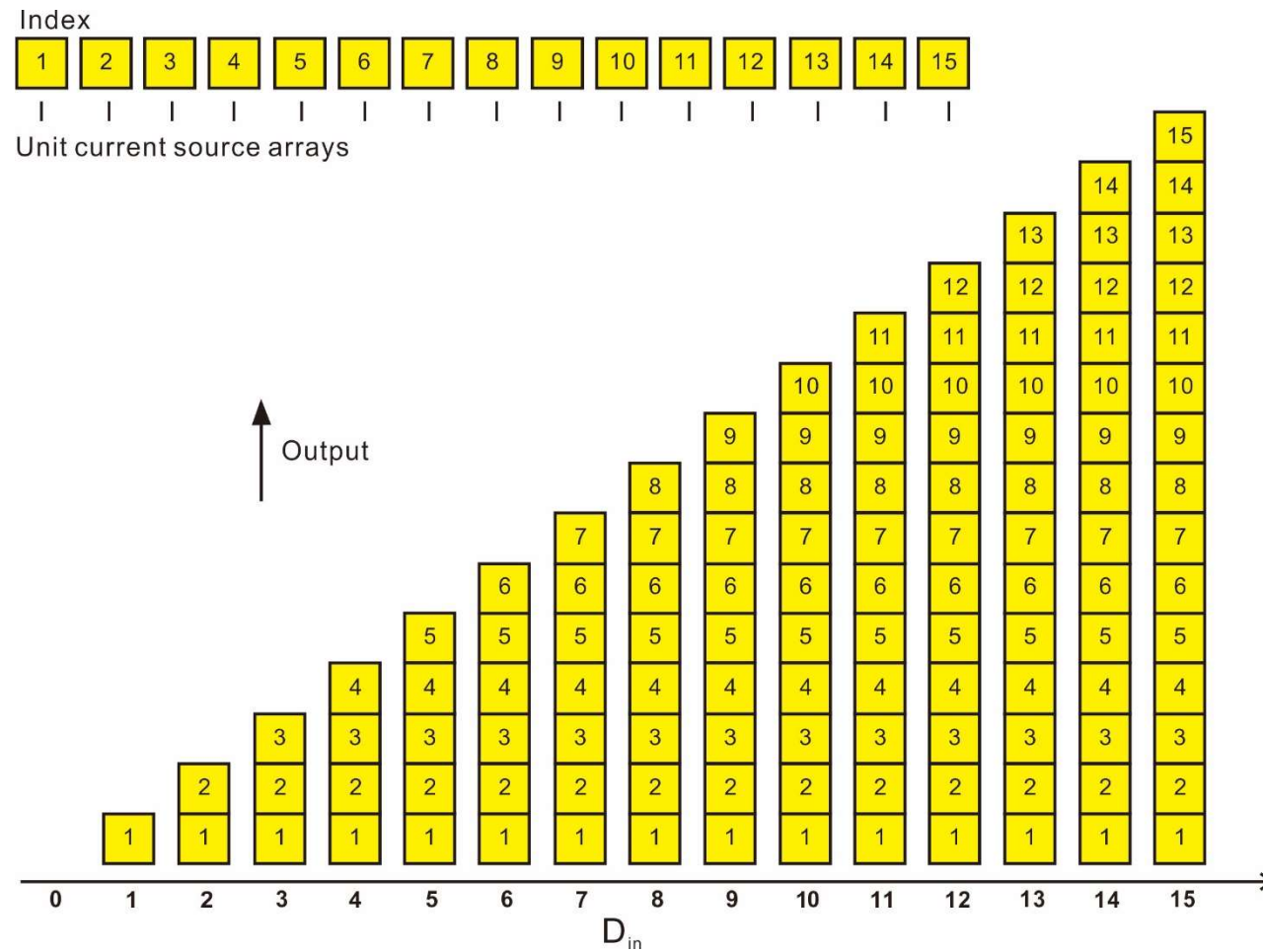


Thermometer Current Steering DAC

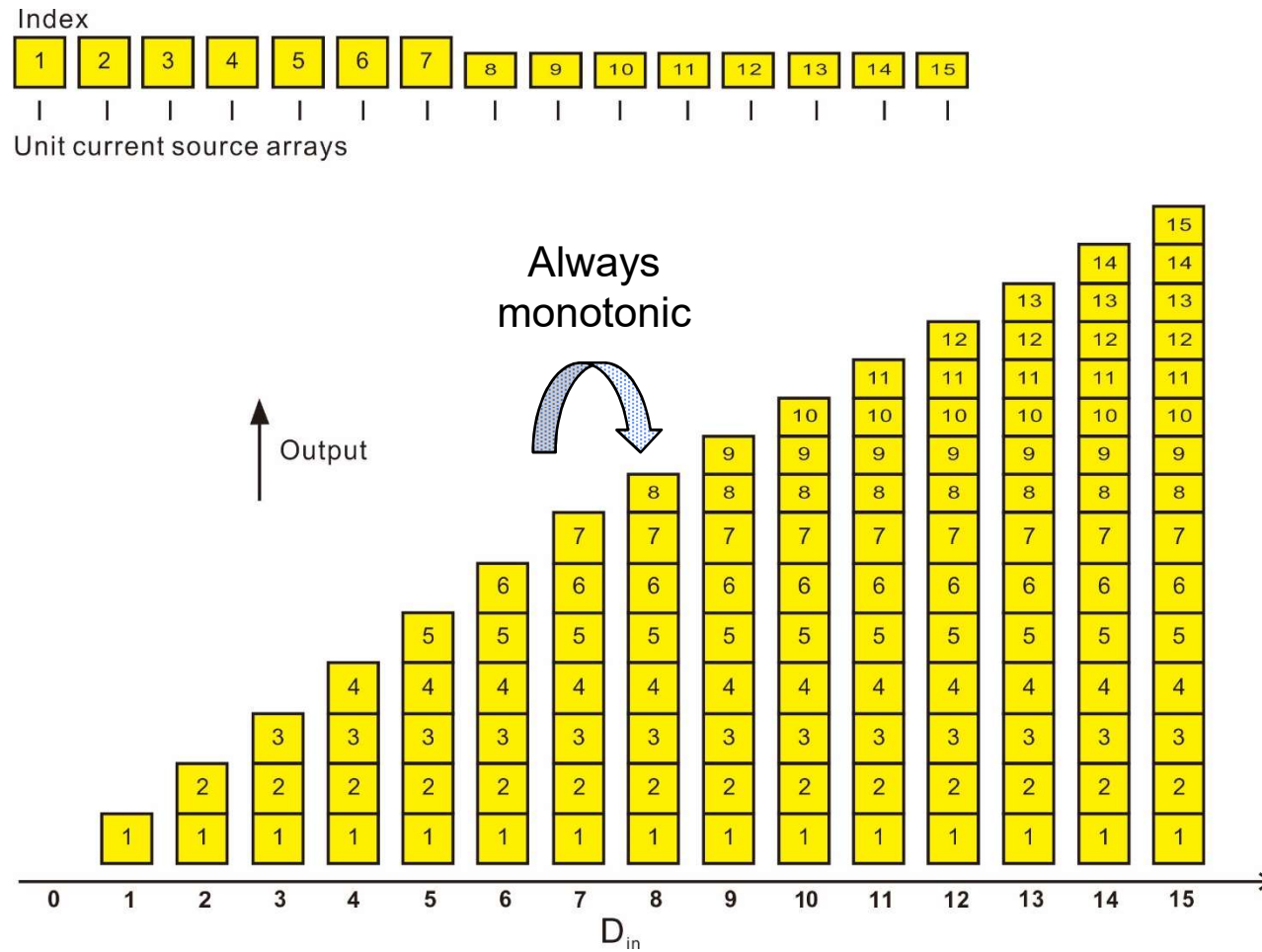


- Inherently monotonic 😊
- Need large encoder with $2^B - 1$ outputs ☹
 - Impractical for large B

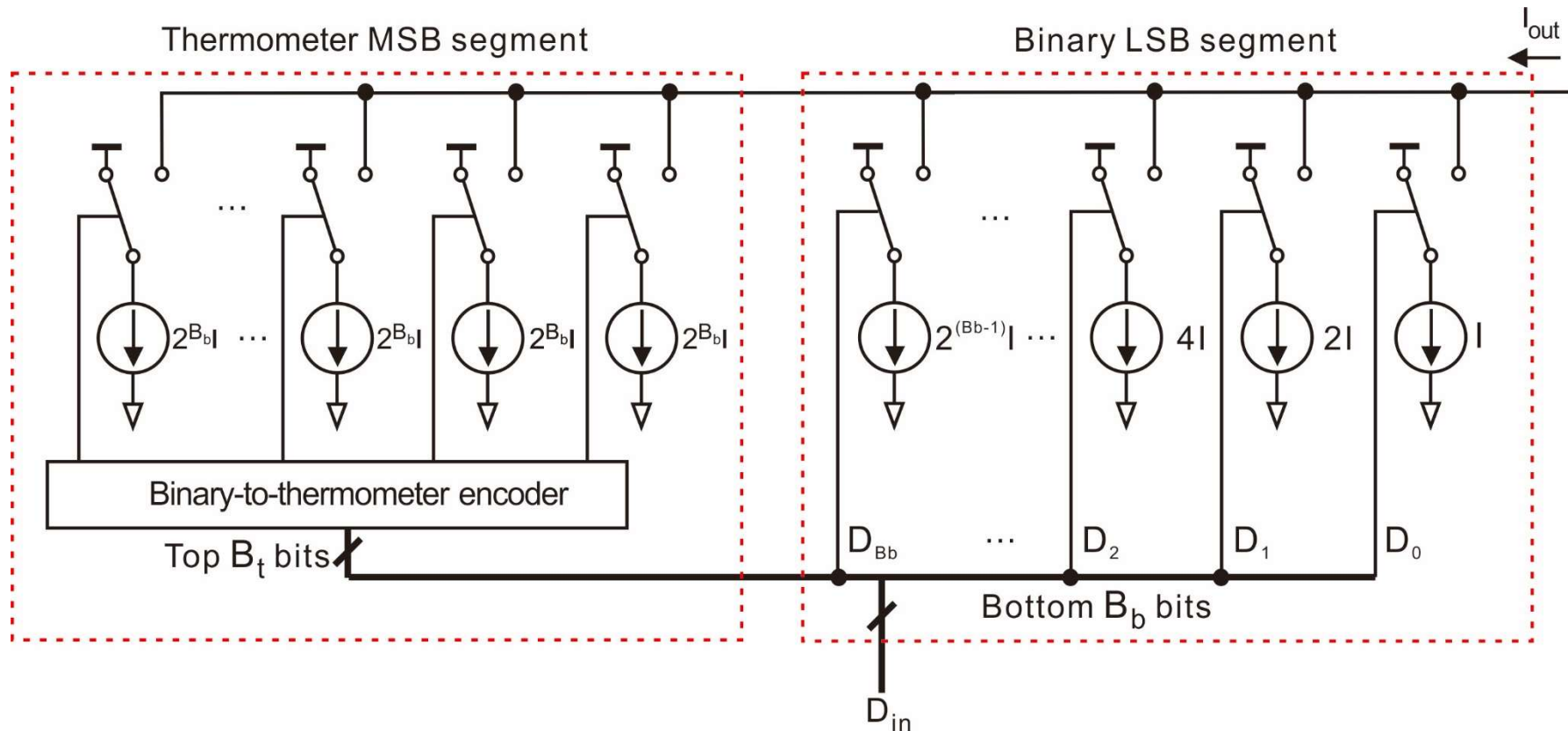
Thermometer DAC Principle



Monotonicity Guaranteed

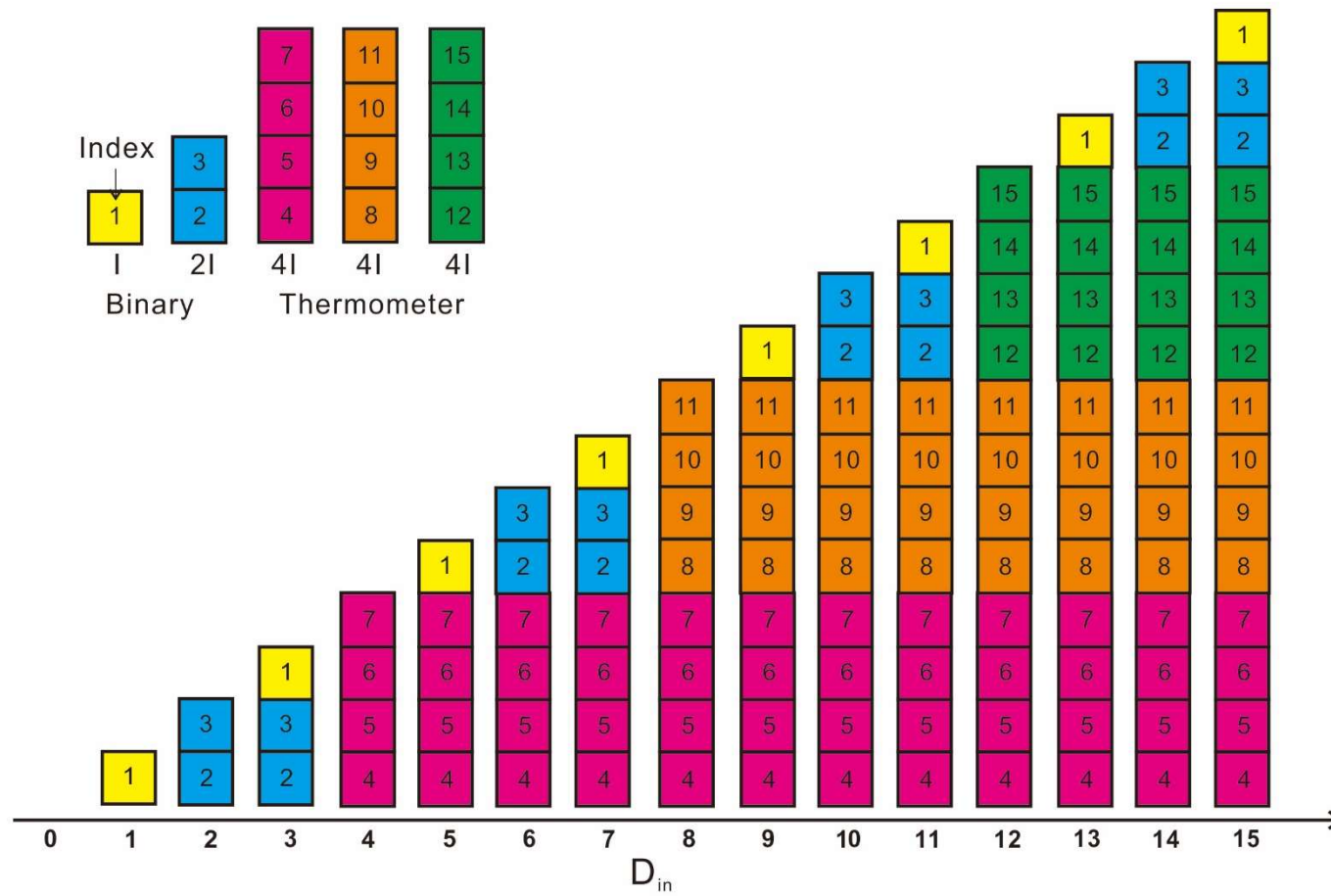


Segmented DAC

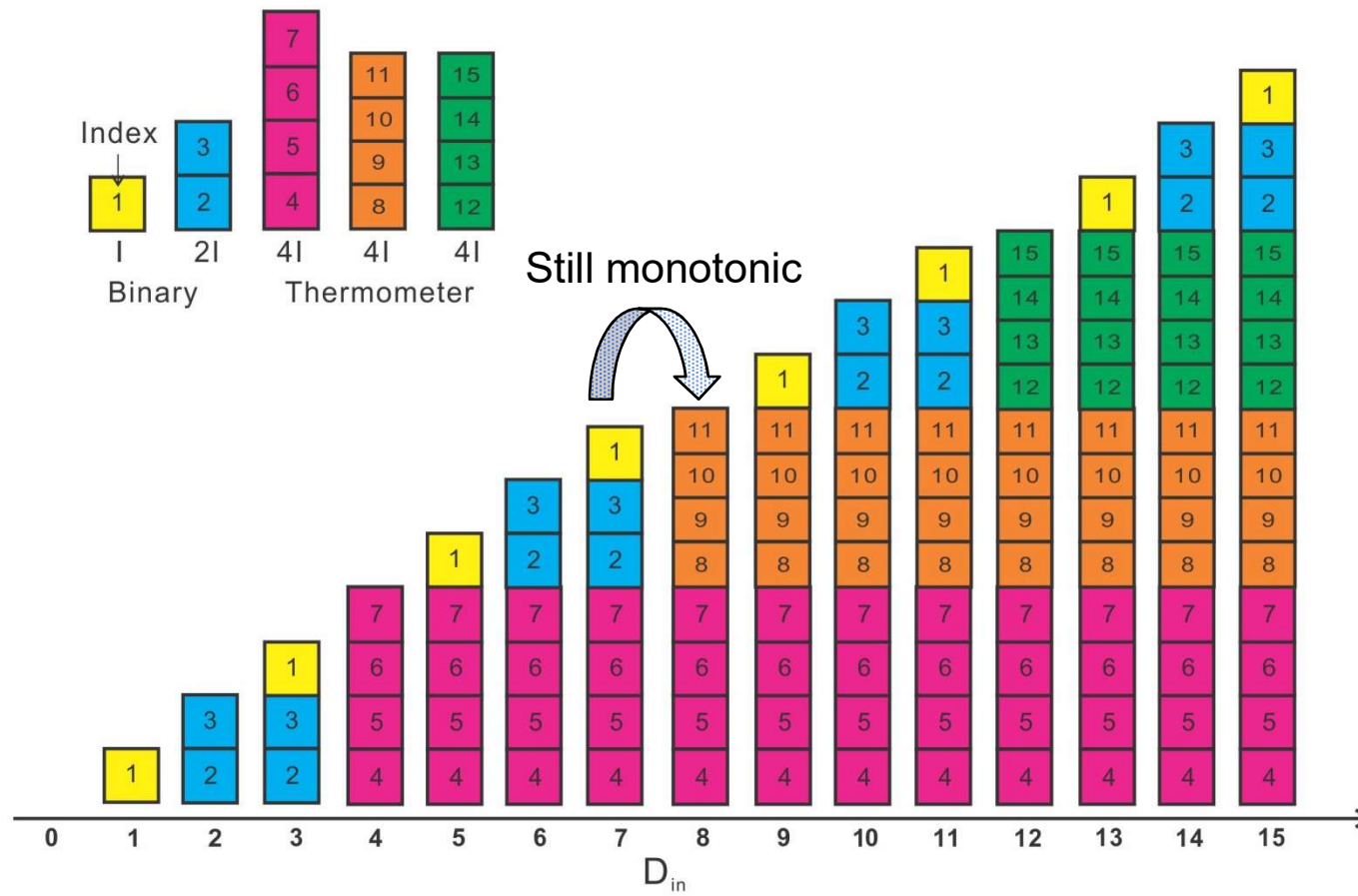


- Binary segment with B_b bits; thermometer segment with $B_t = B - B_b$ bits
- Easier to ensure monotonicity than purely binary DAC ☺
- Smaller encoder than purely thermometer DAC ☺

Segmented DAC (2-2)



Easier to Ensure Monotonicity



Overview

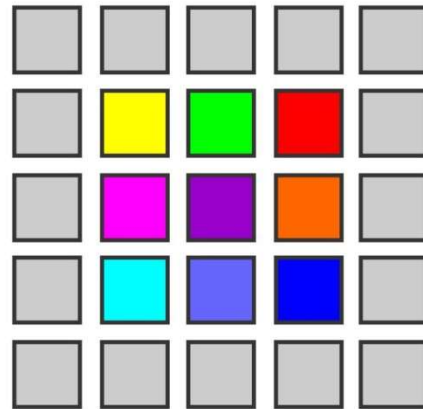
- Topologies
- Current steering DAC architectures
- Static errors
 - Mismatch
 - Finite output impedance
- Dynamic errors
- Example implementations

Systematic Mismatch

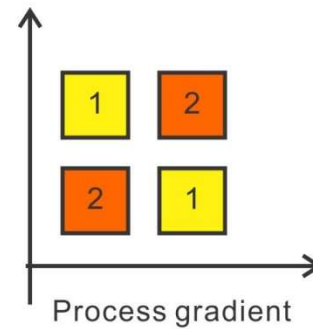
- DNL and INL due to unit element mismatch
- Systematic mismatches
 - IR drop due to contact and wiring resistance
 - Process artifact
 - Gradient
 - Well proximity effect
 - Length of diffusion (shallow trench isolation)
 - Metal coverage
- Systematic mismatches can be mitigated through proper layout
 - Add dummy cell
 - Common centroid layout

Layout Techniques to Reduce Gradient Errors

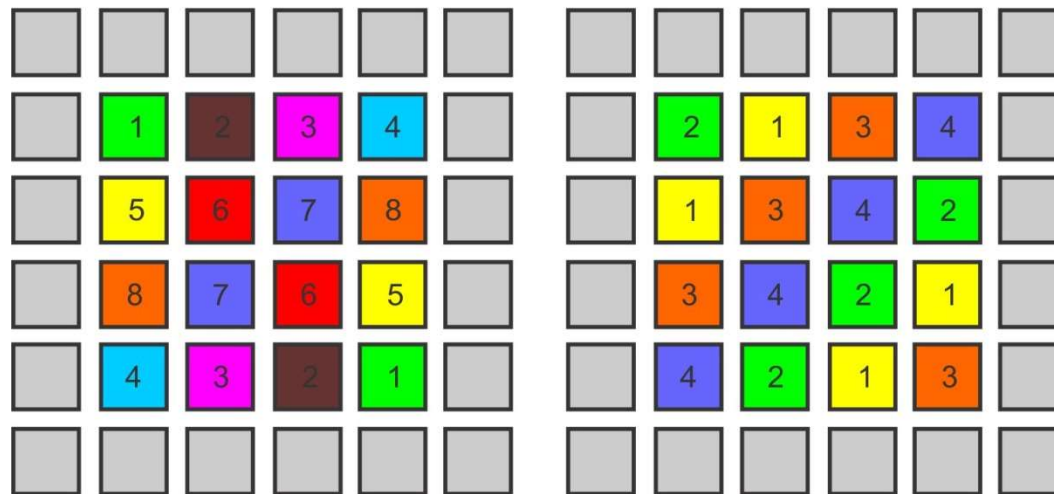
Dummy cell



Common centroid layout



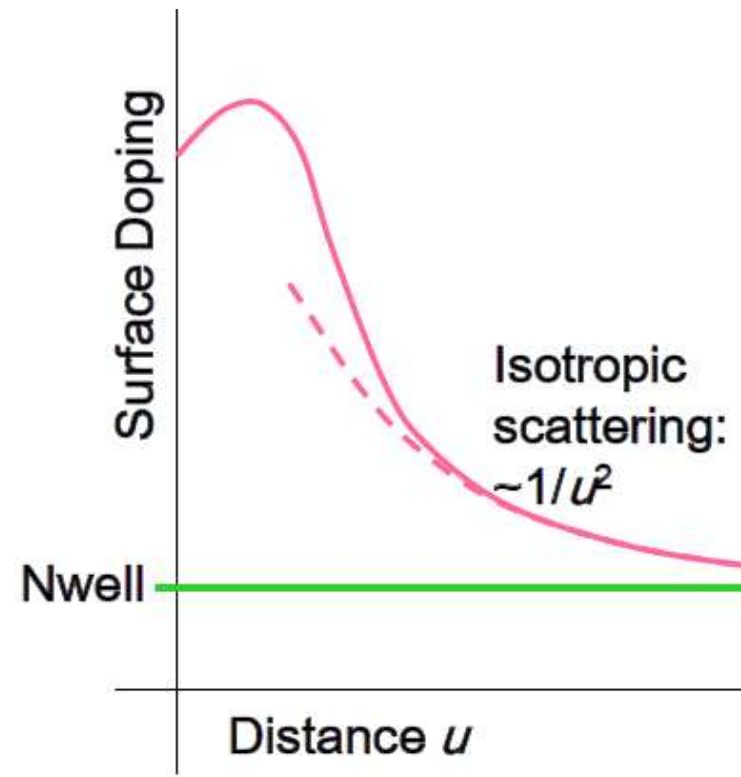
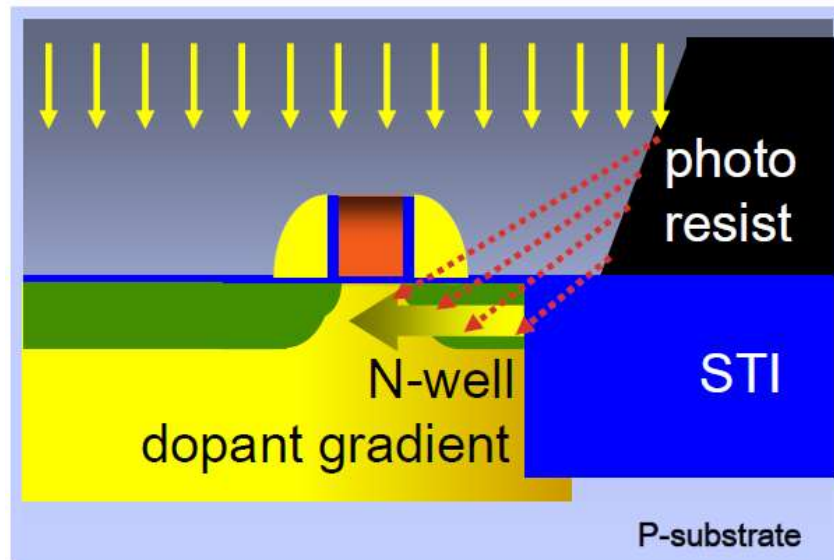
More complicated



Gradient Errors

- In modern CMOS process, parametric gradients are generally well below 1 %/cm (1ppm/ μm). They are smaller than other nonidealities.
 - Most mixed-signal blocks are spread out over less than 2mm
 - Most IP-blocks are smaller than $100 \times 100 \mu\text{m}^2$
- Use common centroid layouts only for very big components

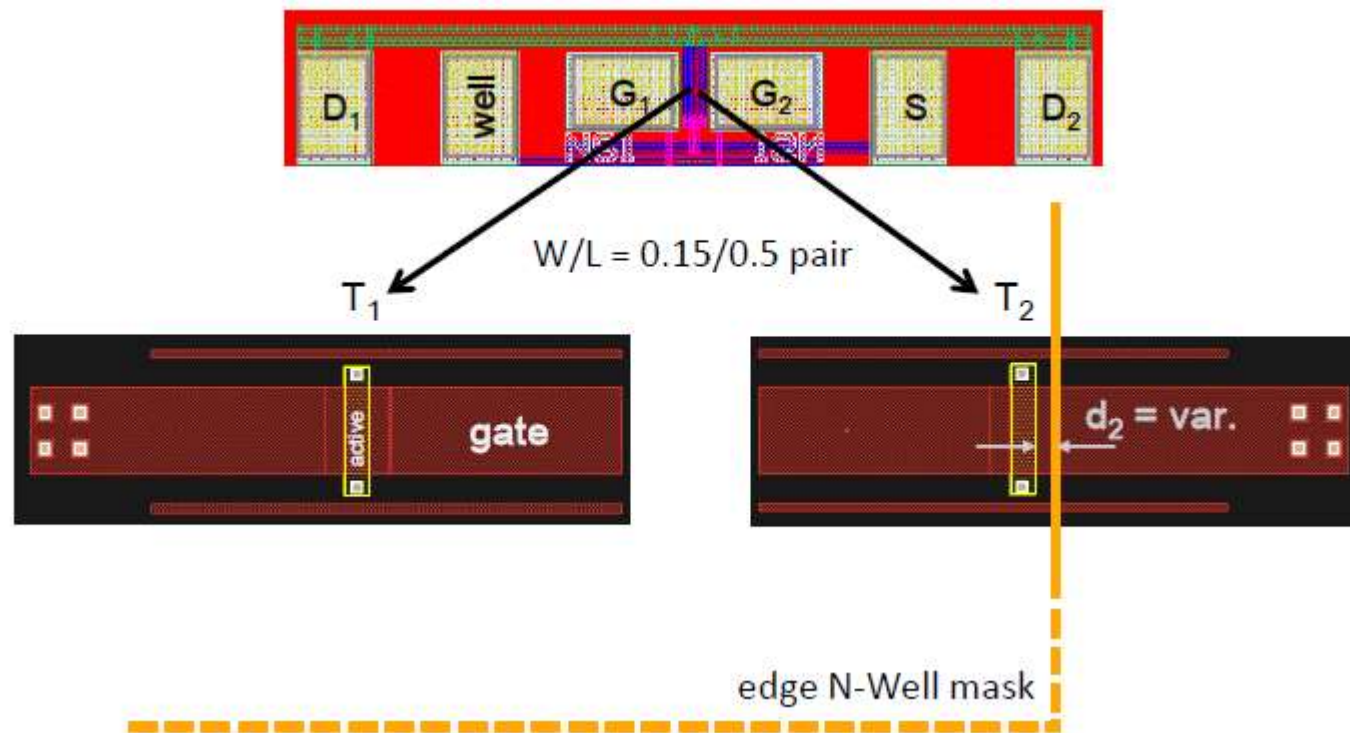
Well Proximity Effect



From H. Tuinhout, CICC 13', Education Session 2.

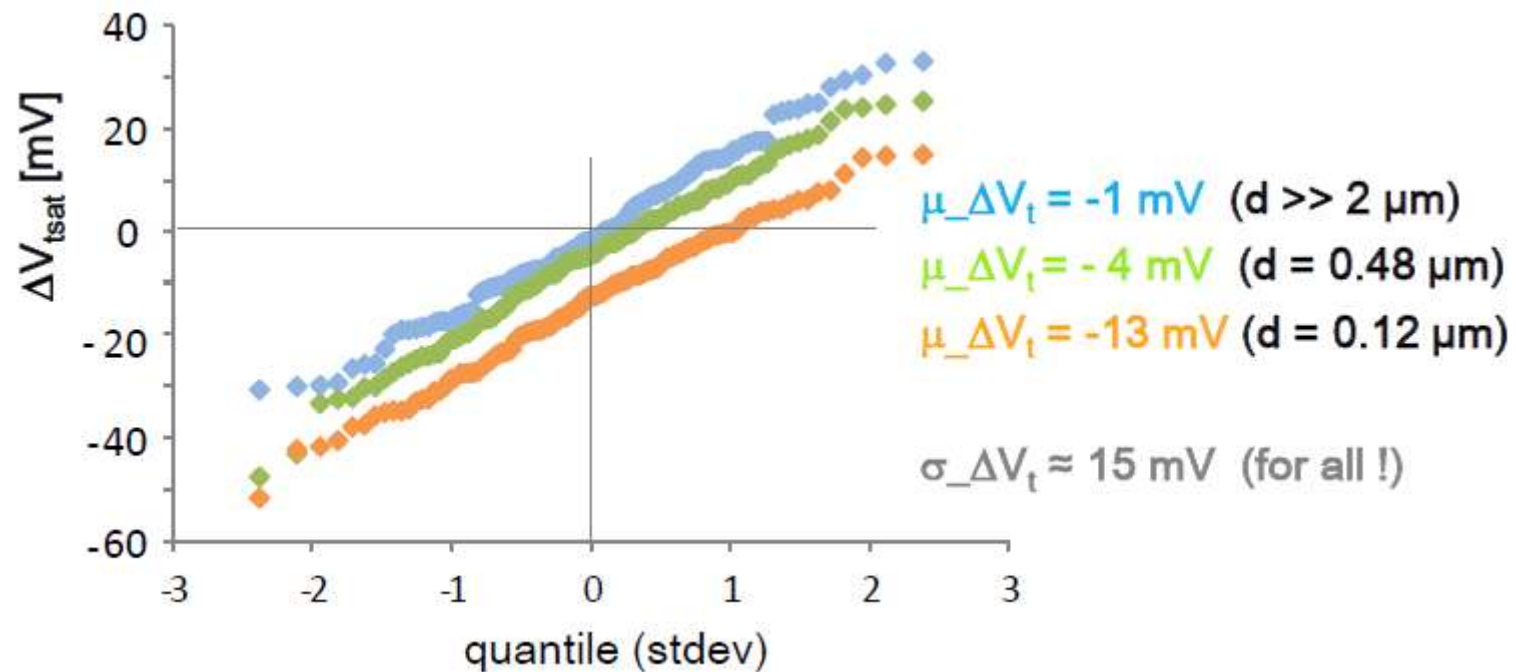
see: T.B. Hook et al.,
IEEE Trans. on Electron Devices,
Vol. 50, No. 9, pp. 1946-1951, 2003.

WPE Effect Study (1)



From H. Tuinhout, CICC 13', Education Session 2.

WPE Effect Study (2)

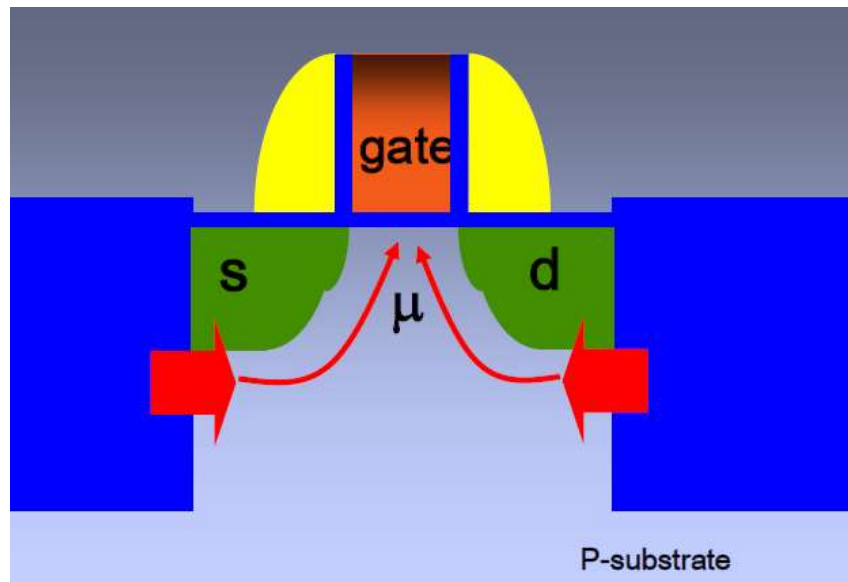


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Well Proximity Effect (WPE)

- WPE depends on:
 - Distance of well edge to active region
 - Well implant dose density and energy
 - Oxide thickness and channel doping profile of the devices
- WPE impacts:
 - V_T variation (up to 30mV per well edge)
- Method to alleviate WPE problem
 - Ensure well-to-active-region distance $>1\mu\text{m}$

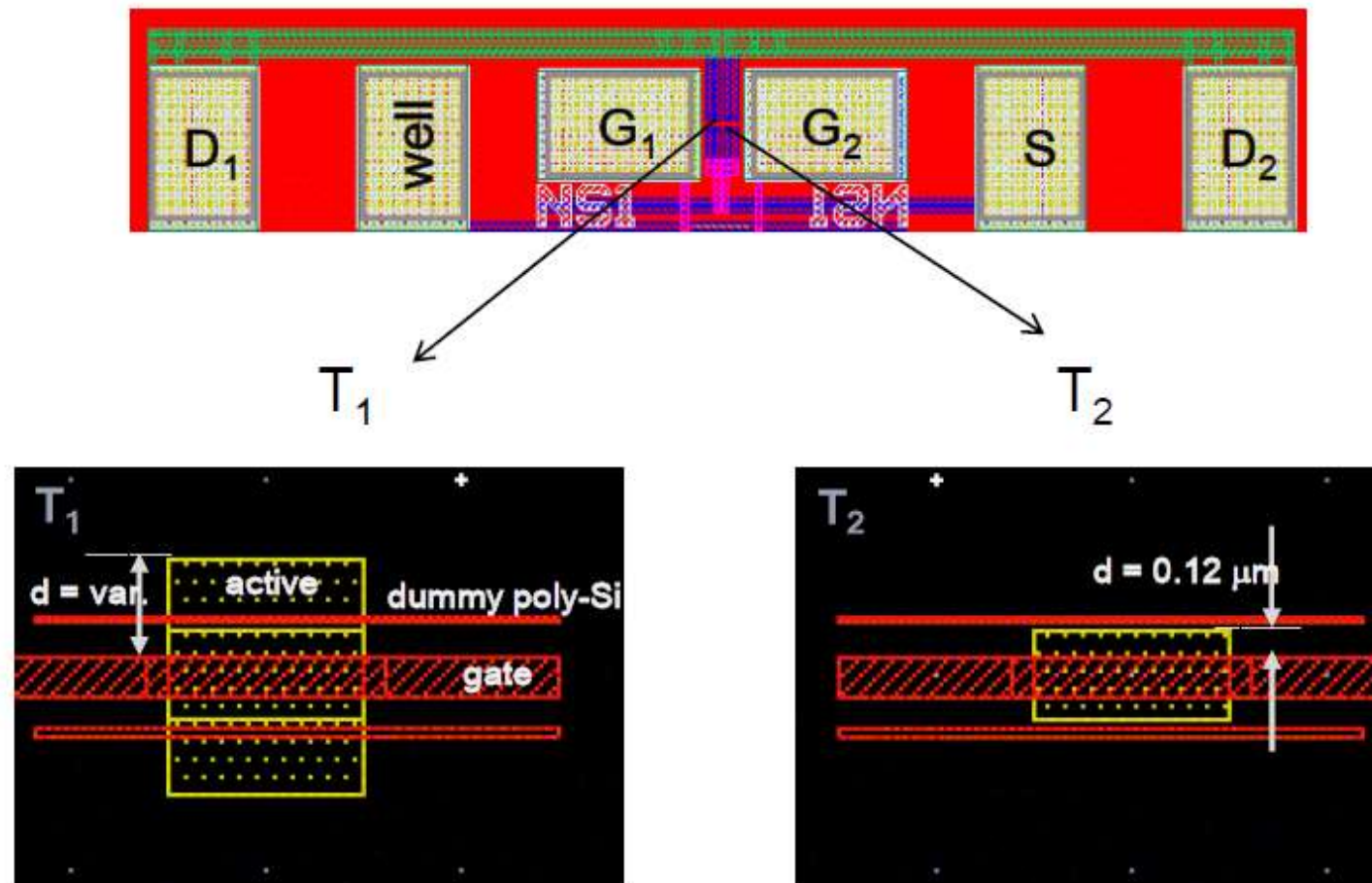
Shallow Trench Isolation (STI)



- STI produces compressive mechanical stress
 - Increases hole μ , reduces electron μ
 - Stress effect proportional to inverse of distance (length of diffusion)
 - Makes device current dependent on layout environment

From H. Tuinhout, CICC 13', Education Session 2.

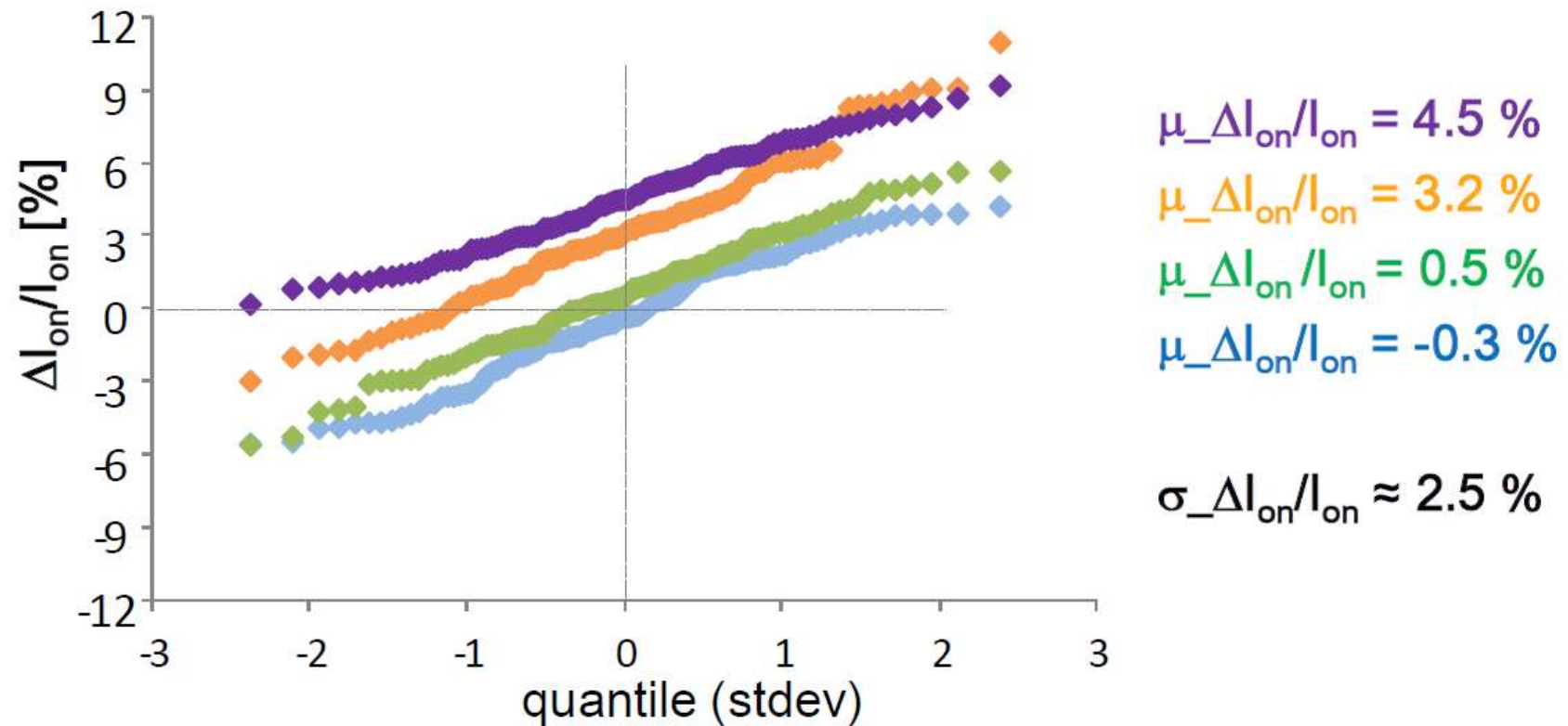
STI Effect Study (1)



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STI Effect Study (2)

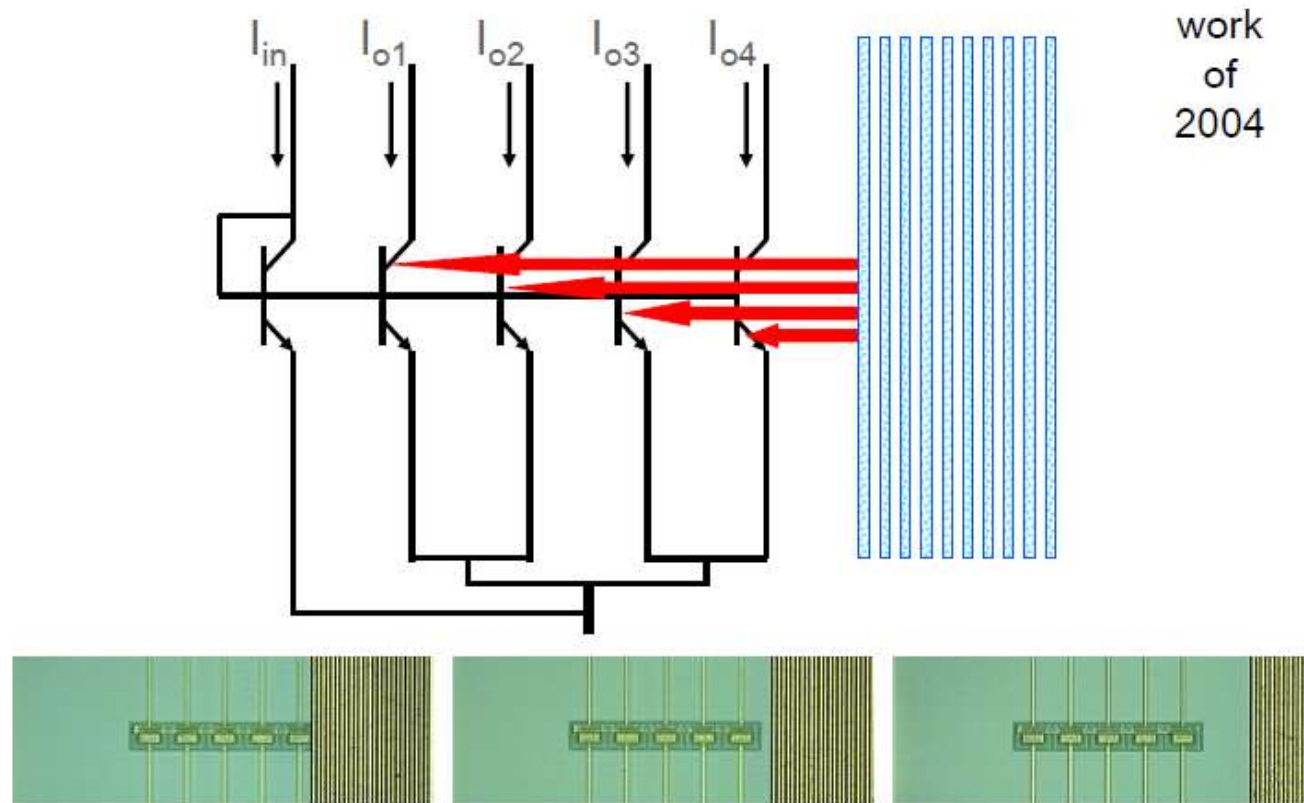
$d = 0.12$ (= reference), 0.14 , 0.27 and $0.5 \mu\text{m}$



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Metal Induced Stress (1)

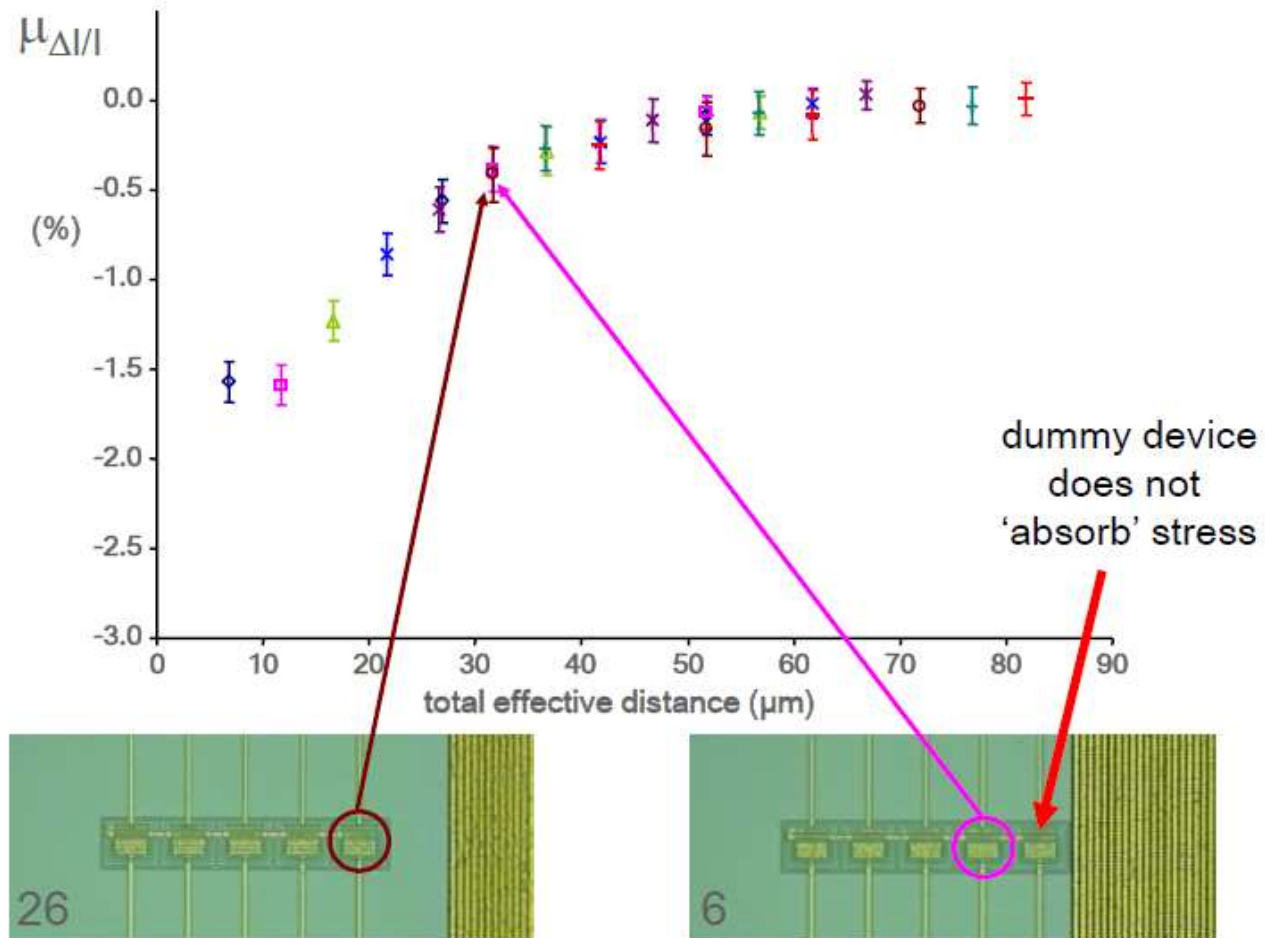
a useful set of experiment with current mirrors and distant stress sources



distance between T_4 and asymmetry is varied from minimum to 36 microns

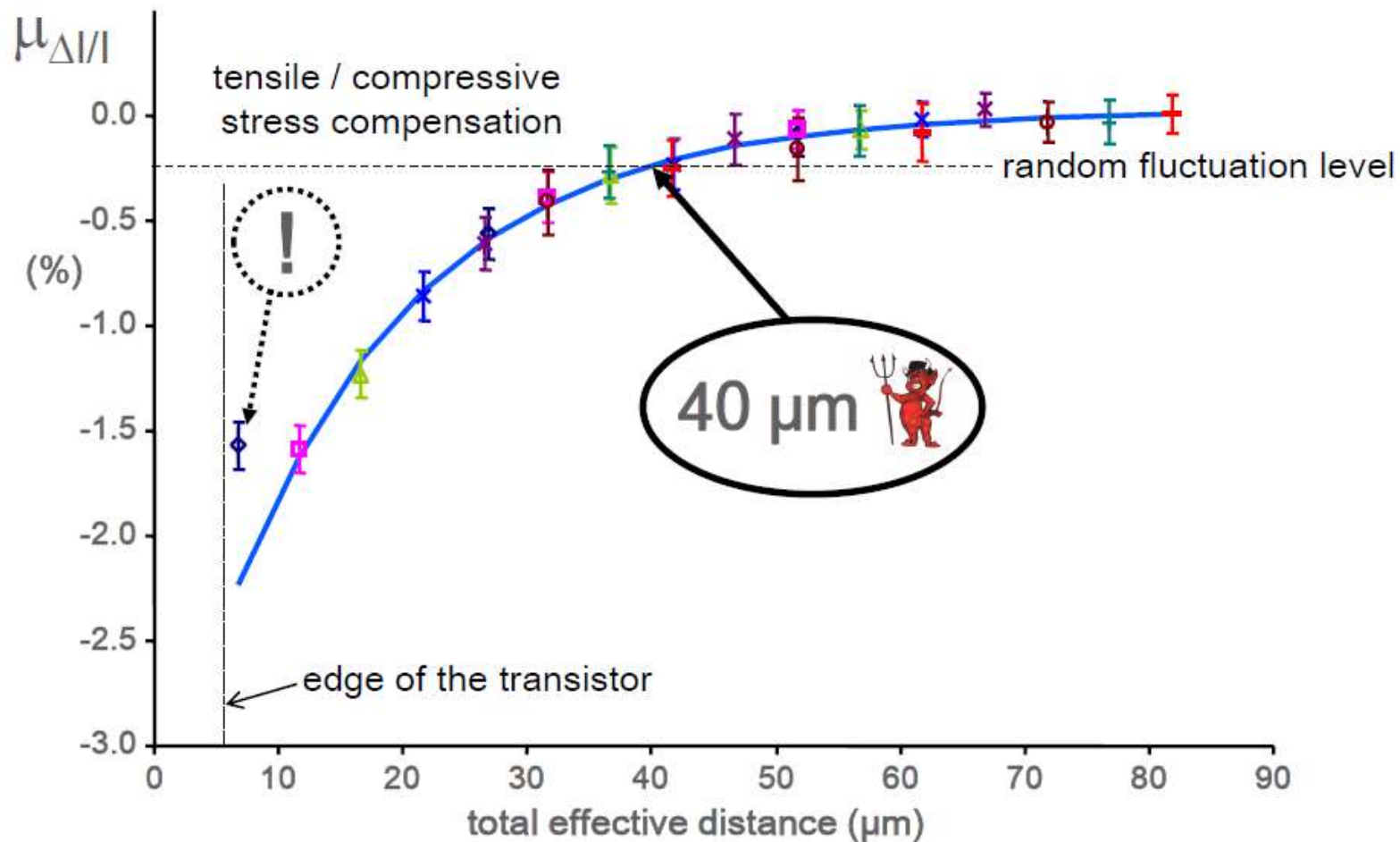
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Metal Induced Stress (2)



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Metal Induced Stress (3)



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Random Mismatch

- Random mismatches are more fundamental and cannot be improved through layout
 - Random number of dopants (V_t variation)
 - Random etching (W & L variation)
- For a well-designed DAC, its nonlinearity should be limited by random mismatch, not systematic mismatch
- What is the distribution of the random mismatch:
 - Normal distribution (central limit theorem)
 - For a unit current source $I[k]$
 - Mean: I
 - Standard deviation: σ
 - We can write $I[k] = I(1 + \Delta I/I)$, then the percentage mismatch $\Delta I/I$ has:
 - Mean: 0
 - Standard deviation: $\sigma_{\Delta I/I}$

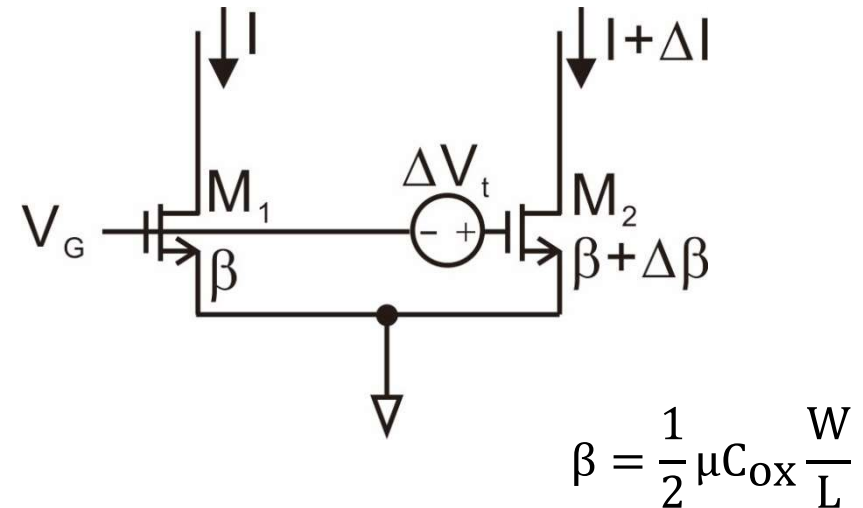
Mismatch in MOS Current Sources

$$\frac{\Delta I}{I} \approx \frac{g_m \Delta V_t}{I} + \frac{\Delta \beta}{\beta}$$

$$\Rightarrow \sigma_{\Delta I/I} = \sqrt{\left(\frac{g_m}{I}\right)^2 \sigma_{\Delta V_t}^2 + \sigma_{\Delta \beta/\beta}^2}$$

$$\sigma_{\Delta V_t} = \frac{A_{V_t}}{\sqrt{WL}}$$

$$\sigma_{\Delta \beta/\beta} = \frac{A_\beta}{\sqrt{WL}}$$



- Example

- $W=10\mu\text{m}$, $L=0.1\mu\text{m}$, $g_m/I=10\text{S/A}$, $A_{V_t}=5\text{mV}\cdot\mu\text{m}$, $A_\beta=1\%\cdot\mu\text{m}$

$$\sigma_{\Delta I/I} = \sigma_u = \sqrt{(10 \cdot 0.005)^2 + (0.01)^2} = \sqrt{(5\%)^2 + (1\%)^2} \approx 5.1\%$$

- Mismatch is usually dominated by V_t mismatch.

How to Reduce Random Mismatch

- Use larger transistors with bigger W and bigger L
 - Larger area ☹️
 - Larger parasitics ☹️
- Reduce g_m/I
 - Bias in strong inversion $\rightarrow$ large V_{ov} $\rightarrow$ reduced output swing ☹️
- Reduce A_{vt} and A_β
 - $A_{vt} \propto t_{ox} \cdot \sqrt[4]{N_A}$
 - Better process $\rightarrow$ increased cost ☹️

DAC INL/DNL Analysis

- Question to ask:
 - Given σ_u , how large are DAC INL and DNL?
 - It depends on different architectures
 - Thermometer DAC
 - Binary-weighted DAC
 - Segmented DAC

Thermometer DAC DNL Analysis

- For thermometer DAC, the step size for code k equals the magnitude of the k-th current source.

$$\text{DNL}[k] = \frac{\text{Step}[k] - \text{Step}_{\text{avg}}}{\text{Step}_{\text{avg}}} \approx \frac{I_k - I}{I} \approx \frac{\Delta I}{I}$$

$$\Rightarrow \sigma_{\text{DNL}[k]} = \sigma_{\Delta I/I} = \sigma_u$$

Thermometer DAC DNL Yield

- Question to ask:
 - We have a 8-bit thermometer DAC with $\sigma_u = 1\%$. We want the DAC DNL to be within ± 0.03 LSB. What is the yield after fabrication?
- Analysis
 - ± 0.03 LSB = $\pm 3\% = \pm 3\sigma_u$
 - For one code, its width falls in between $\pm 3\sigma$ is 99.73%, according to normal distribution.
 - We want all $2^8=256$ codes satisfy the requirement, thus, the yield is: $(99.73\%)^{256} \approx 50\%$
- How about a 12-bit DAC?
 - All 4096 codes satisfy the requirement $\rightarrow$ yield = $(99.73\%)^{4096} \approx 0$
 - Let us relax the DNL requirement to ± 0.05 LSB $\rightarrow$ yield $\approx 99\%$
 - A big change, why?

Thermometer DAC INL Analysis

$$\text{INL}[k] = \frac{I_{\text{out}}[k] - I_0[k]}{\text{Step}_{\text{avg}}} = \frac{\sum_{j=1}^k I_j - \frac{k}{N} \sum_{j=1}^N I_j}{\frac{1}{N} \sum_{j=1}^N I_j} = N \frac{\sum_{j=1}^k I_j}{\sum_{j=1}^N I_j} - k = N \frac{X}{Y} - k$$

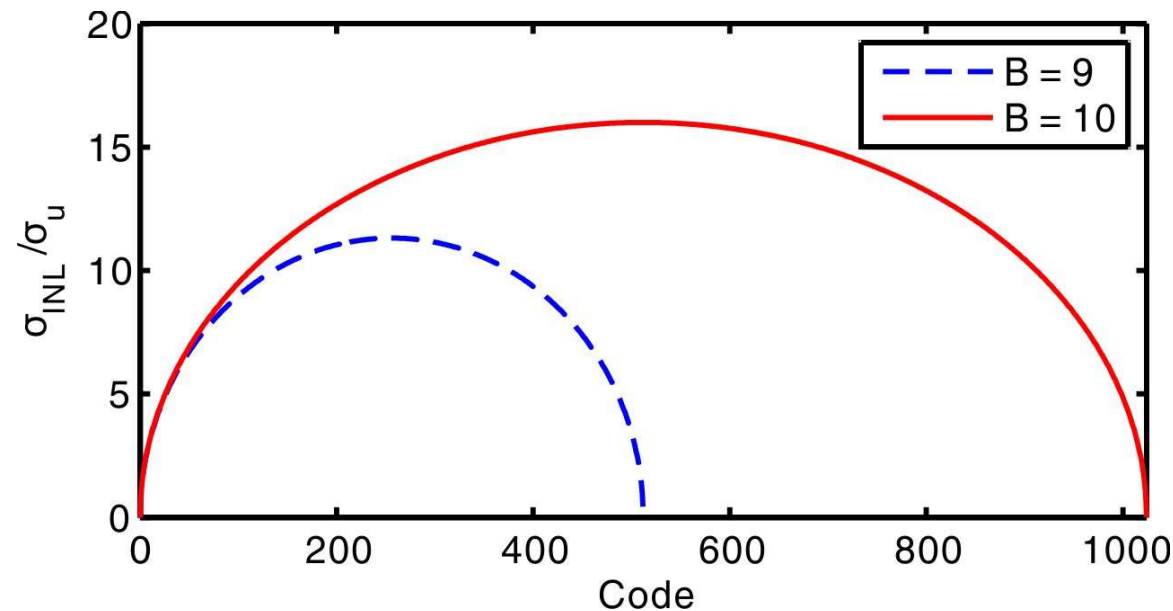
$$\Rightarrow \text{var}[\text{INL}[k]] = \text{var} \left[N \frac{X}{Y} - k \right] = N^2 \text{var} \left[\frac{X}{Y} \right]$$

$$\frac{X}{Y} = \frac{\mu_X + \Delta X}{\mu_Y + \Delta Y} \approx \frac{\mu_X}{\mu_Y} + \frac{\Delta X}{\mu_Y} - \frac{\mu_X \Delta Y}{\mu_Y^2}$$

$$\text{var} \left[\frac{X}{Y} \right] \approx \left(\frac{\mu_X}{\mu_Y} \right)^2 \left(\frac{\sigma_X^2}{\mu_X^2} + \frac{\sigma_Y^2}{\mu_Y^2} - 2 \frac{\text{cov}(X, Y)}{\mu_X \mu_Y} \right)$$

$$\text{var}[\text{INL}[k]] \approx k(1 - k/N) \sigma_u^2 \quad \sigma_{\text{INL}}[k] \approx \sigma_u \sqrt{k(1 - k/N)}$$

Thermometer DAC σ_{INL} Plot

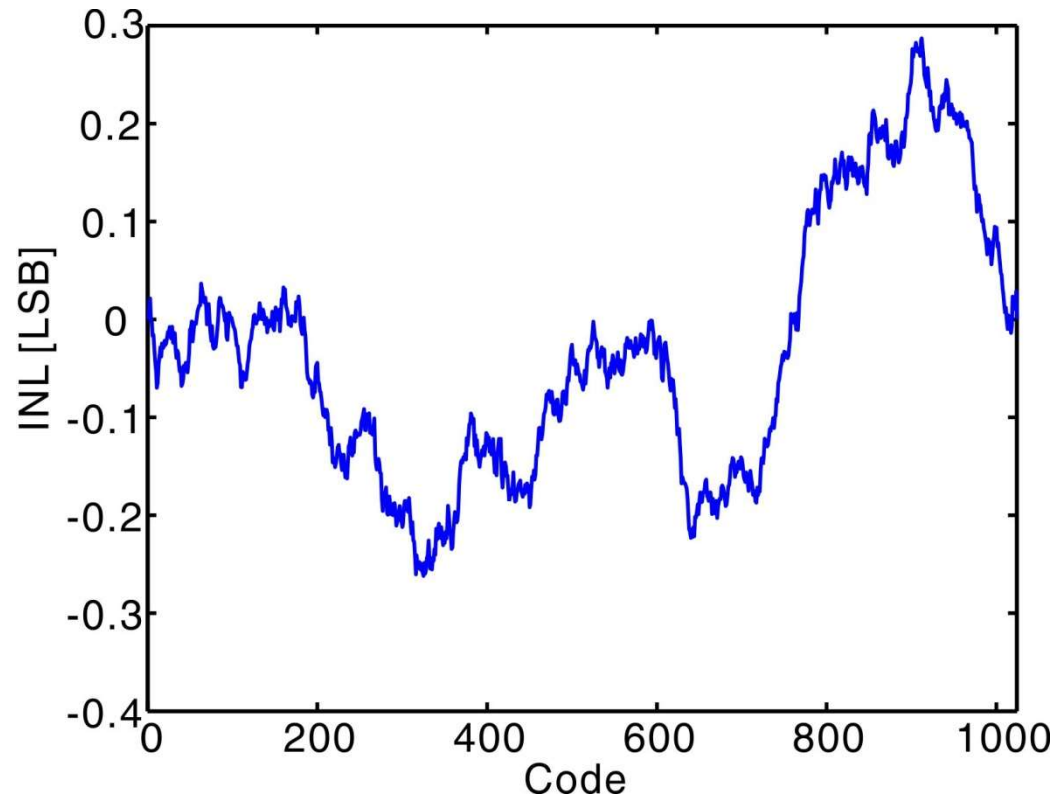


- σ_{INL} is maximum at mid-scale ($k=N/2=2^{B-1}$)

$$\sigma_{\text{INL}}[N/2] \approx \sigma_u \sqrt{\frac{N}{2} \left(1 - \frac{N/2}{N}\right)} = \frac{\sigma_u}{2} \sqrt{N} \approx \sigma_u 2^{B/2-1}$$

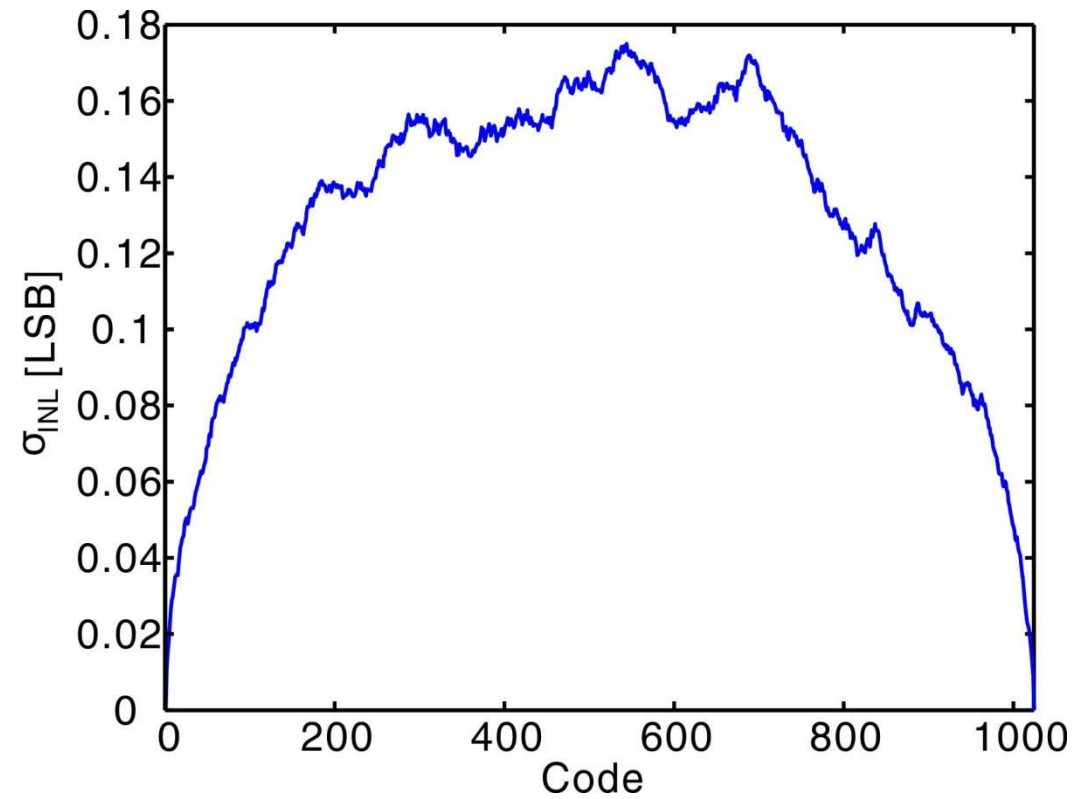
- This is a “Brownian bridge:” summation of random DNLs
- Unlike σ_{DNL} , σ_{INL} grows as B increases.

Thermometer DAC INL



- 1 run for a 10-b DAC with $\sigma_u=1\%$

Thermometer DAC σ_{INL} Plot



- 100 runs, plot σ_{INL}
- Match well with theory

Thermometer DAC INL Yield

- Question to ask:
 - We have a 8-bit thermometer DAC with $\sigma_u = 1\%$. We want the DAC INL to be within ± 0.03 LSB. What is the yield after fabrication?
 - It turns out that this is not easy to solve analytically
 - Unlike $\{DNL[k]\}$, $\{INL[k]\}$ are not independent
- In general, DAC designers will rely on numerical Monte Carlo simulations to estimate the DNL/INL yield instead of solving analytical solutions
 - This method is conceptually very simple
 - It works for all types of DACs (thermometer, binary, and segmented)
 - Yet a basic understanding will be very helpful for finding errors in the simulation

Binary Weighted DAC DNL Analysis

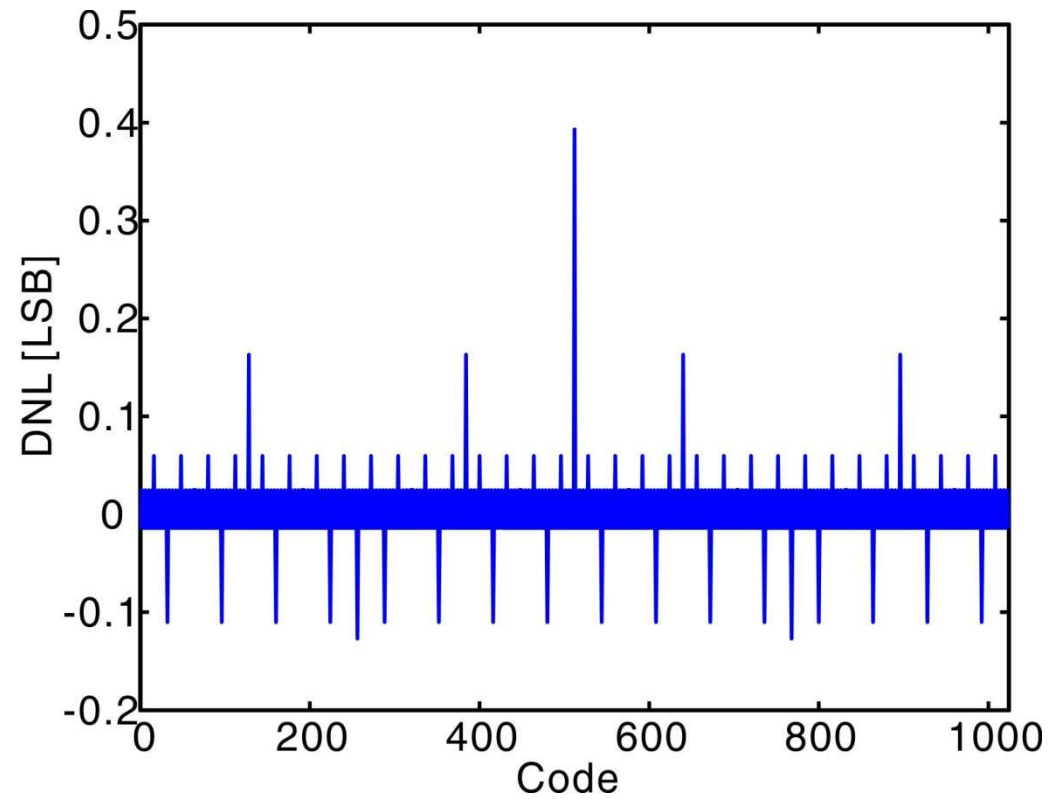
- $\{DNL[k]\}$ are different and depend on transition
- Worst case: 01111111 ... $\rightarrow$ 1000000 ...
 - Turning off all LSBs and turn on the MSB

$$\text{Step}[2^{B-1}] = \sum_{i=2^{B-1}}^{2^B-1} I_i - \sum_{j=1}^{2^{B-1}-1} I_j$$
$$DNL[2^{B-1}] = \frac{\text{Step}[2^{B-1}] - \text{Step}_{\text{avg}}}{\text{Step}_{\text{avg}}} \approx \sum_{i=2^{B-1}}^{2^B-1} \frac{\Delta I_i}{I} - \sum_{j=1}^{2^{B-1}-1} \frac{\Delta I_j}{I}$$

$$\sigma_{DNL, \text{worst}}^2 = (2^{B-1}) \sigma_u^2 + (2^{B-1} - 1) \sigma_u^2 = (2^B - 1) \sigma_u^2$$

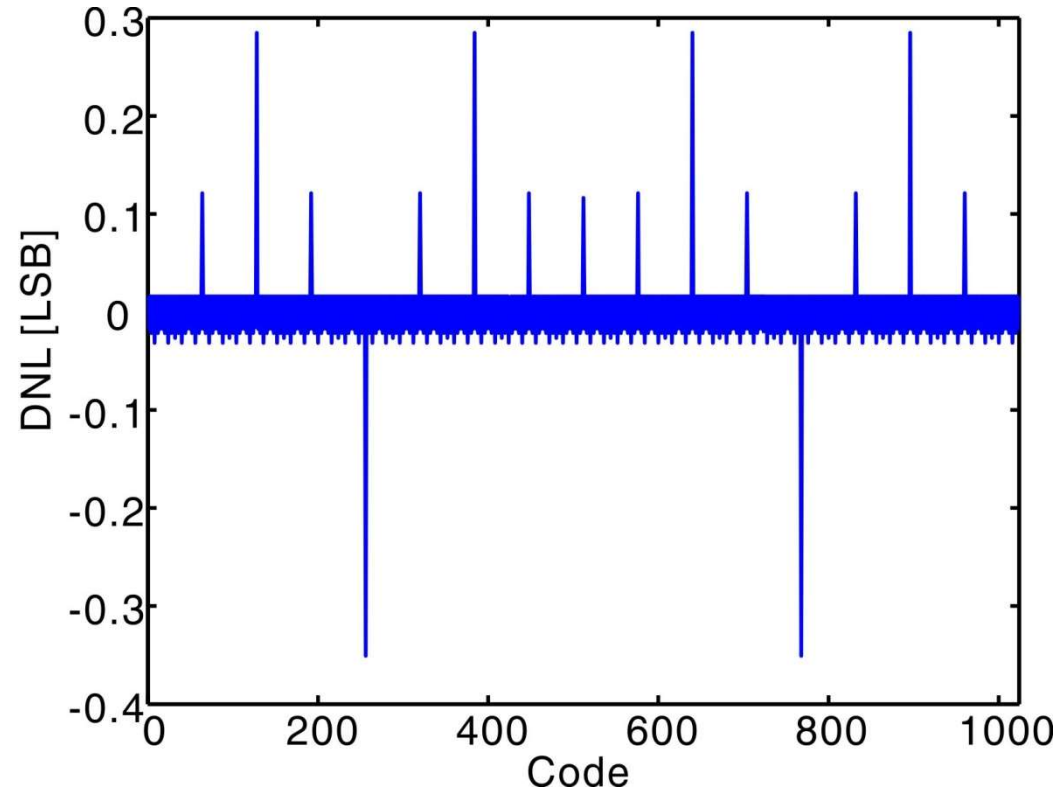
- For example, $B = 10$, $\sigma_u = 1\% \rightarrow \sigma_{DNL, \text{worst}} = 0.32 \text{ LSB}$
 - Much larger than that of the thermometer DAC ($\sigma_{DNL} = 0.01 \text{ LSB}$).

Binary Weighted DAC DNL



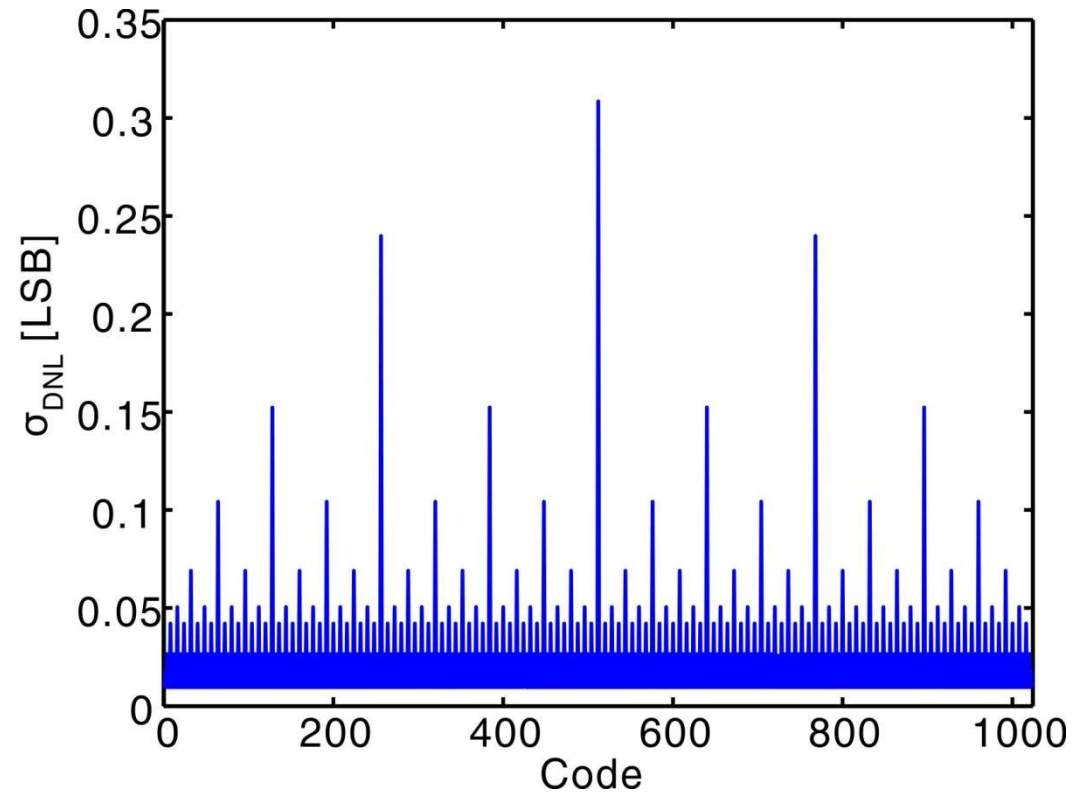
- 1 run for a 10-b DAC with $\sigma_u=1\%$

Binary Weighted DAC DNL



- Another run, worst DNL does not happen at the MSB transition
 - Due to randomness

Binary Weighted DAC σ_{DNL}



- 100 runs, plot σ_{DNL}
- Match well with theory

Binary Weighted DAC INL Analysis

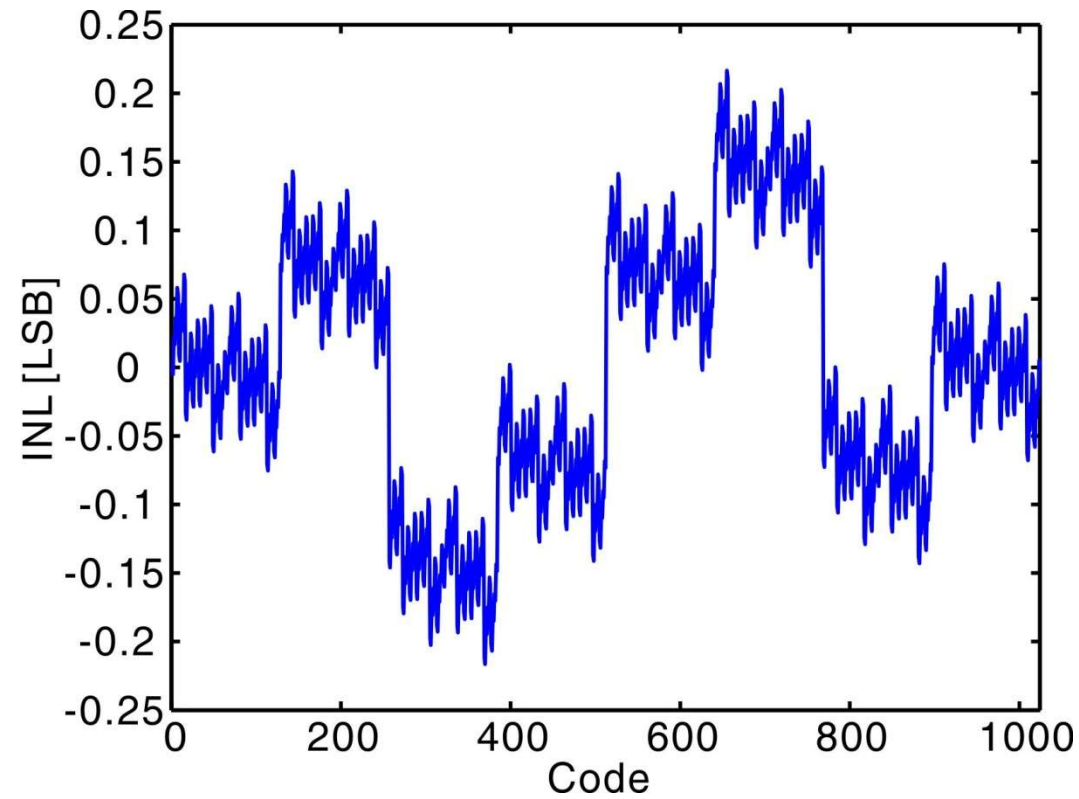
- σ_{INL} is the same as that for a thermometer DAC
 - Why?
 - INL is the overall deviation from the ideal straight line
 - It does not matter how we select the elements

$$\text{INL}[k] = \frac{I_{\text{out}}[k] - I_0[k]}{\text{Step}_{\text{avg}}}$$

$$\sigma_{\text{INL}}[k] \approx \sigma_u \sqrt{k(1 - k/N)}$$

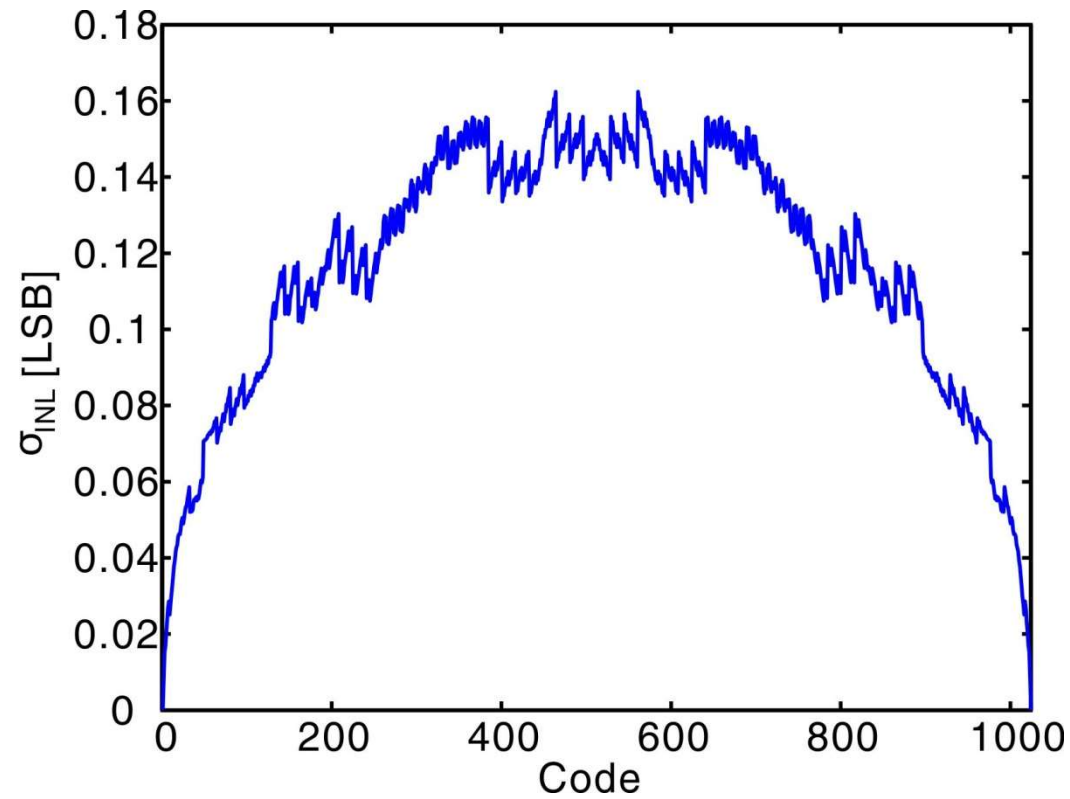
- What is the difference in the INL of a binary weighted DAC?
 - There are large discontinuities due to large DNLs

Binary Weighted DAC INL



- 1 run for a 10-b DAC with $\sigma_u=1\%$

Binary Weighted DAC σ_{INL} Plot

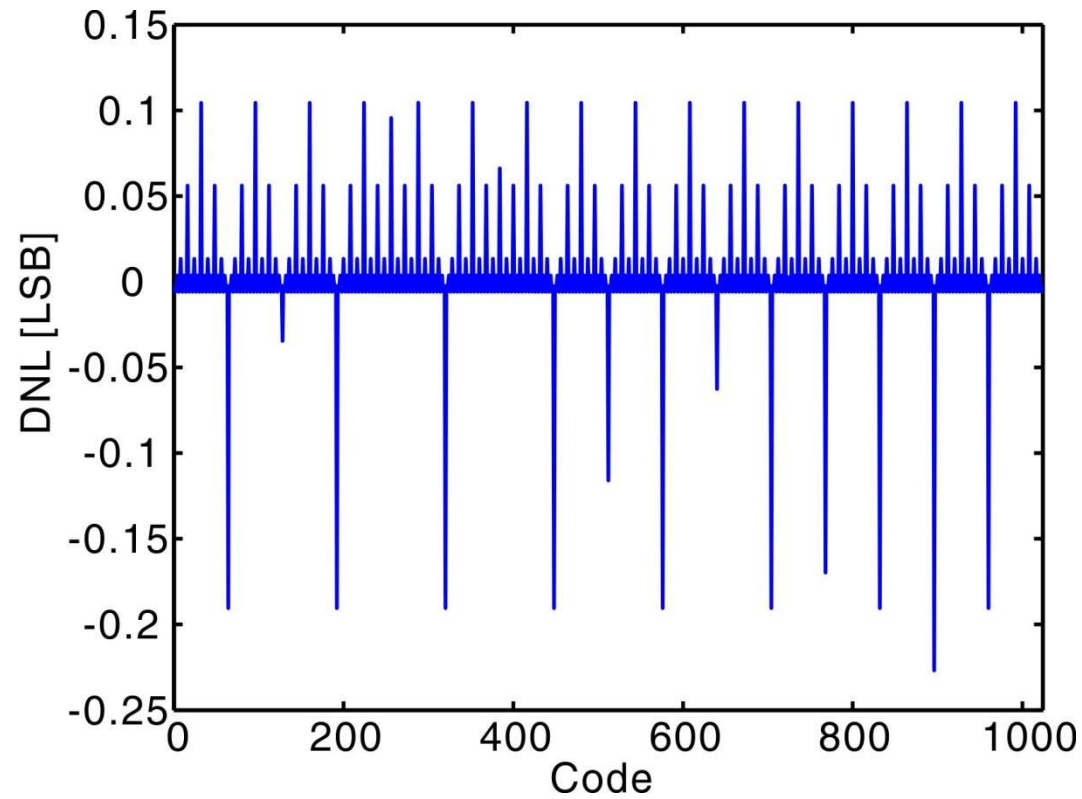


- 100 runs, plot σ_{INL}
- Match well with theory

Segmented DAC DNL/INL Analysis

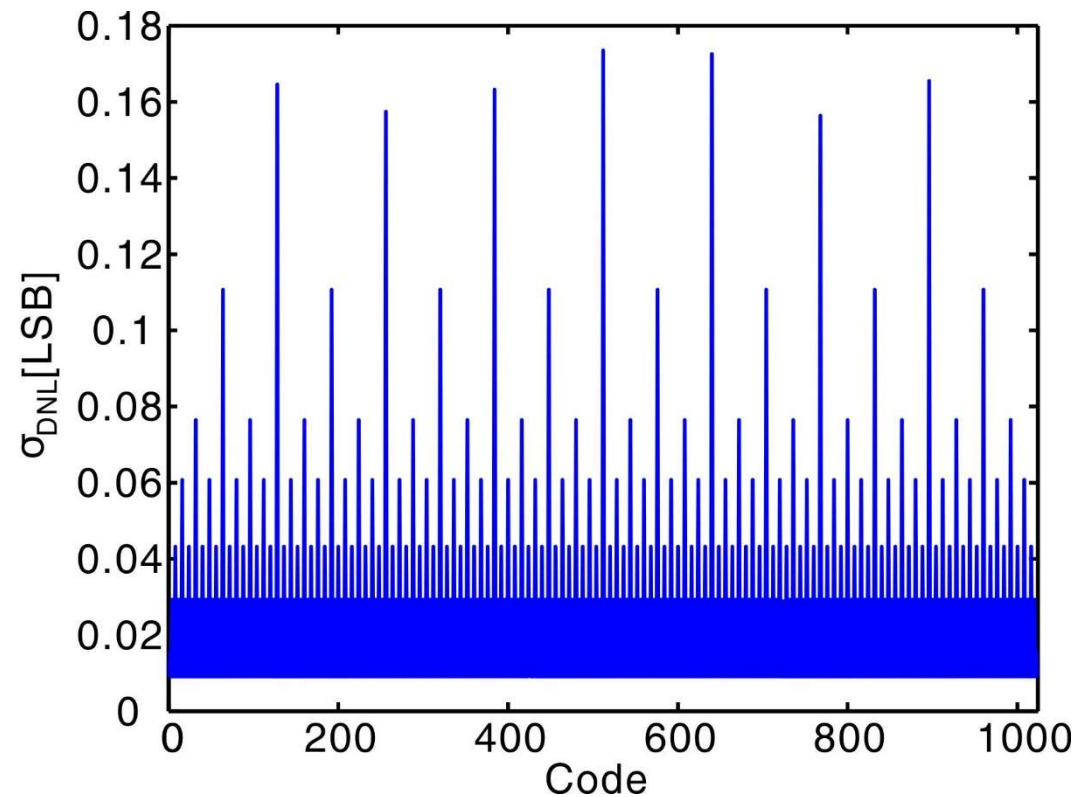
- Worst case DNL
 - When all LSB DAC elements turn off and 1 MSB DAC element turns on
 - Same DNL as a binary weighted DAC with B_b+1 bits
- INL
 - σ_{INL} is the same as that of thermometer DAC

Segmented DAC DNL



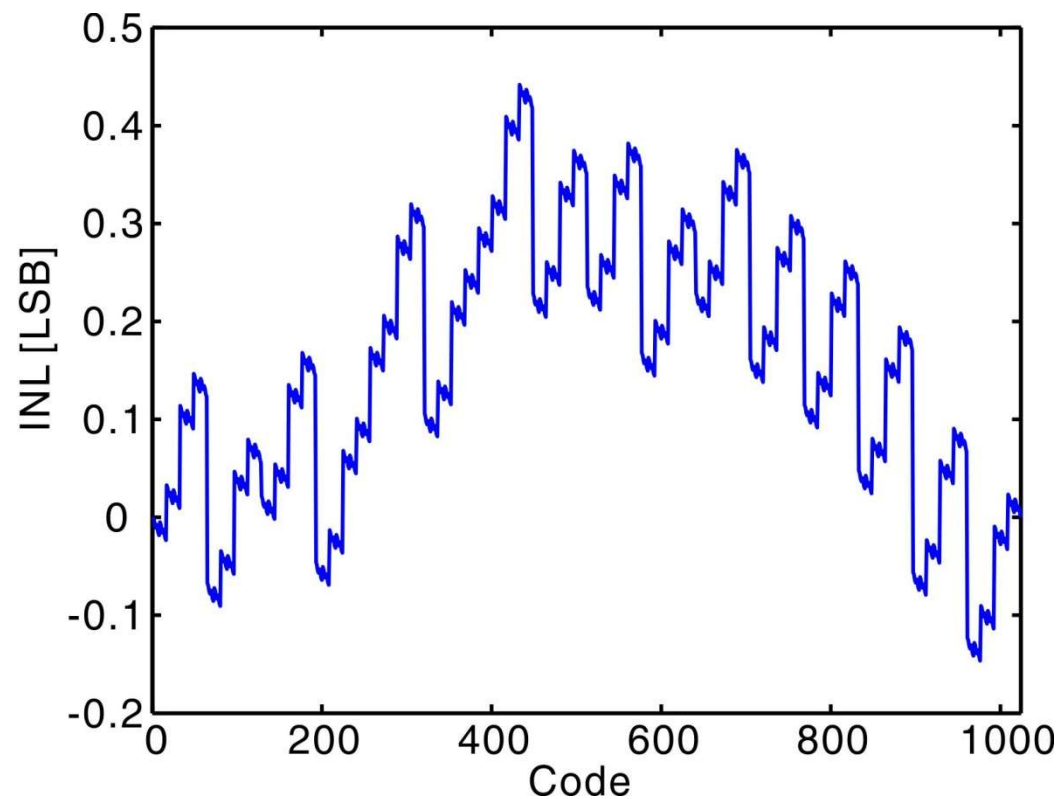
- 1 run for a 10-b segmented DAC (3b MSB and 7b LSB) with $\sigma_u=1\%$

Segmented DAC σ_{DNL} Plot



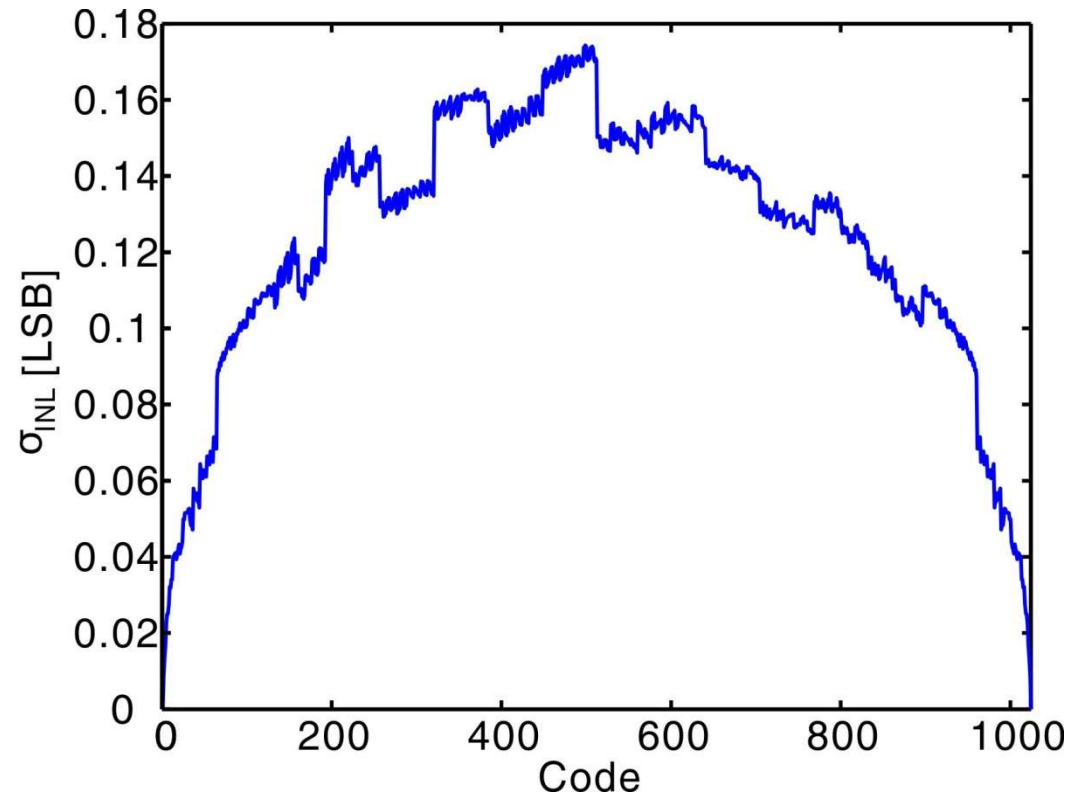
- 100 runs, plot σ_{DNL}
- Match well with theory

Segmented DAC INL



- 1 run for a 10-b segmented DAC (3b MSB and 7b LSB) with $\sigma_u=1\%$

Segmented DAC σ_{INL} Plot



- 100 runs, plot σ_{INL}
- Match well with theory

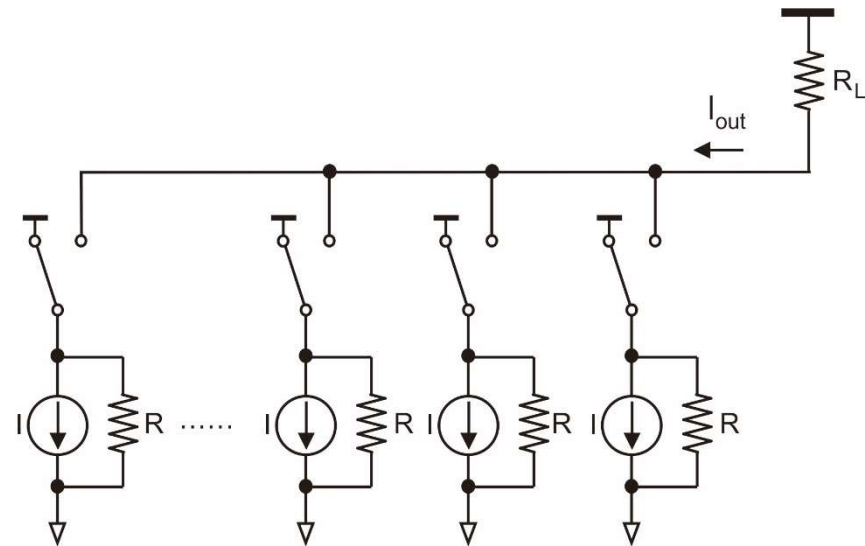
Summary

	Thermometer	Segmented	Binary Weighted
σ_{INL}	$\sigma_{\text{INL}}[k] \approx \sigma_u \sqrt{k(1 - k/N)}$		
σ_{DNL} (worst case)	σ_u	$2^{(B_b+1)/2} \sigma_u$	$2^{B/2} \sigma_u$
# elements	$2^B - 1$	$B_b + 2^{B_t} - 1$	B

- Segmented DAC in general achieves the best trade-off.
- Most high-performance DACs adopt segmented DAC structure.

Finite Output Impedance

- Current source with finite output resistance R
- The number of closed switches:
 $2^B \times \frac{D_{in}}{FS}$
- $V_{out} = 2^B \times \frac{D_{in}}{FS} \times (R_L || \frac{R}{2^B \times \frac{D_{in}}{FS}})$
- As D_{in} increases, do the code widths get wider or narrower?
- Need to make R large enough
 - Cascode
 - Source degeneration



Overview

- Topologies
- Current steering DAC architectures
- Static errors
- **Dynamic errors**
- Example implementations

Dynamic DAC Errors

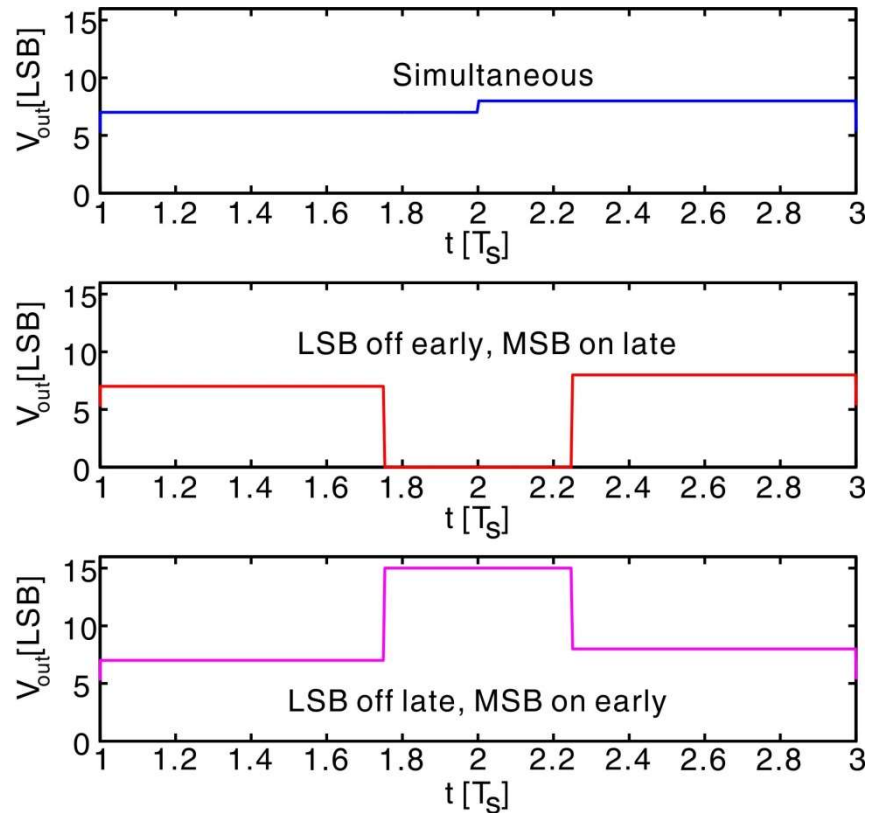
- Dynamic DAC errors are generally hard to model
 - DAC output is continuous in time, and thus, any error at any time instance causes trouble!
 - High-performance DAC design is nontrivial.
- Glitches due to timing errors
 - Current sources do not switch simultaneously
- Clock feedthrough
 - Coupling from switch signals to DAC output
- Finite settling time and slewing
 - Finite RC time constant
 - Signal dependent slewing
- Inter symbol interference
 - Unmatched rising and falling edges

Useful References

- Doris, van Roermund, Leenaerts, Wide-Bandwidth High Dynamic Range D/A Converters, Springer, 2006.
- Bosch, Steyaert, Sansen, Static and Dynamic Performance Limitations for High Speed D/A Converters, Kluwer, 2004.
- Gustavsson, Chapter 12
- M. Albiol, J.L. Gonzalez, E. Alarcon, "Mismatch and dynamic modeling of current sources in current-steering CMOS D/A converters," IEEE TCAS I, pp. 159-169, Jan. 2004
- T. Chen and G.G.E. Gielen, "The analysis and improvement of a current-steering DAC's dynamic SFDR," IEEE Trans. Ckts. Syst. I, pp. 3-15, Jan. 2006.

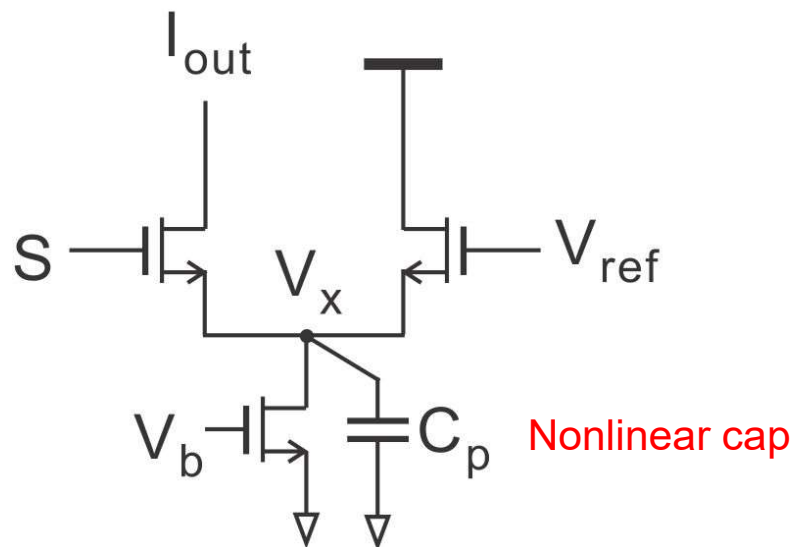
Glitch Impulse Error

- Current sources need to be turned on/off simultaneously
 - Consider binary weighted DAC transition $0111 \rightarrow 1000$
- Worst case glitch impulse area:
 $\propto \Delta t 2^{B-1}$
- LSB area: T_s
- $\Delta t 2^{B-1} \ll T_s \rightarrow \Delta t \ll T_s/2^{B-1}$
- E.g., $T_s = 1\text{ns}$, $B = 10$, $\Delta t < 2\text{ps}$
 - Using a segmented structure will help

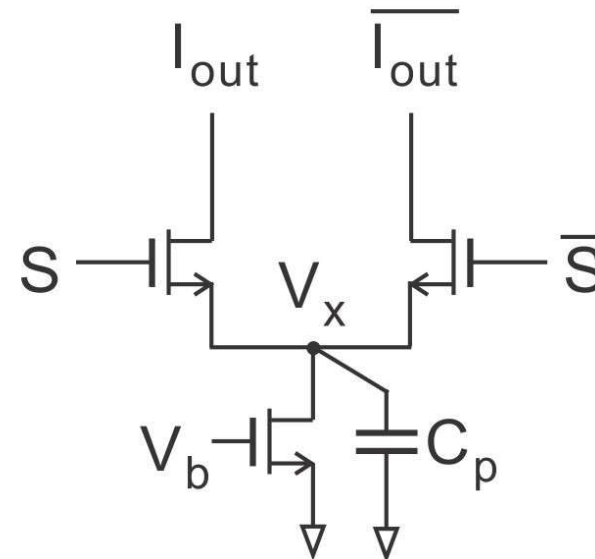


Basic Differential Pair Switch

Single-ended



Differential

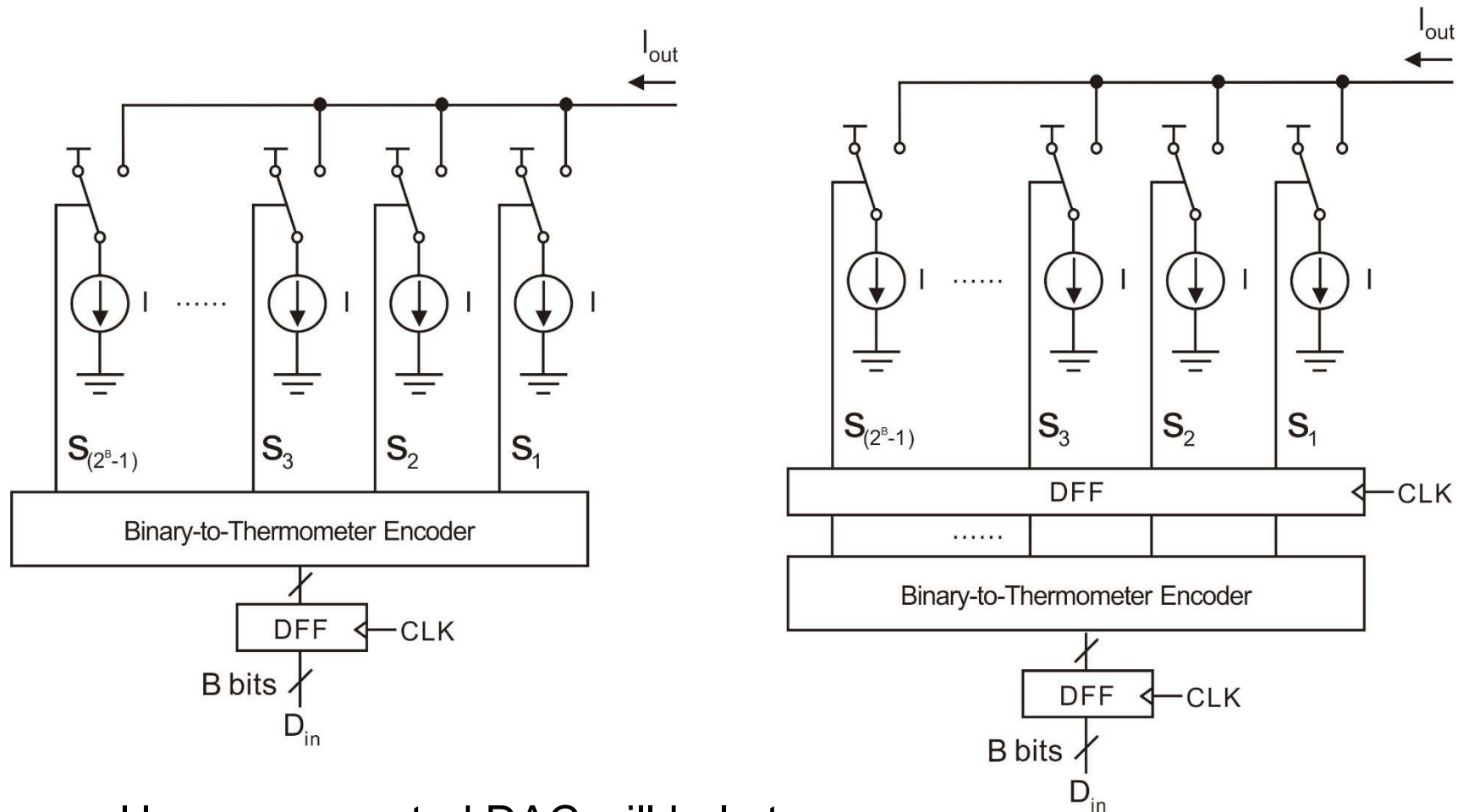


- For high-speed, never turn off the current source
 - Parasitic cap needs to be recharged
- Need to minimize the swing at V_x

Design Techniques

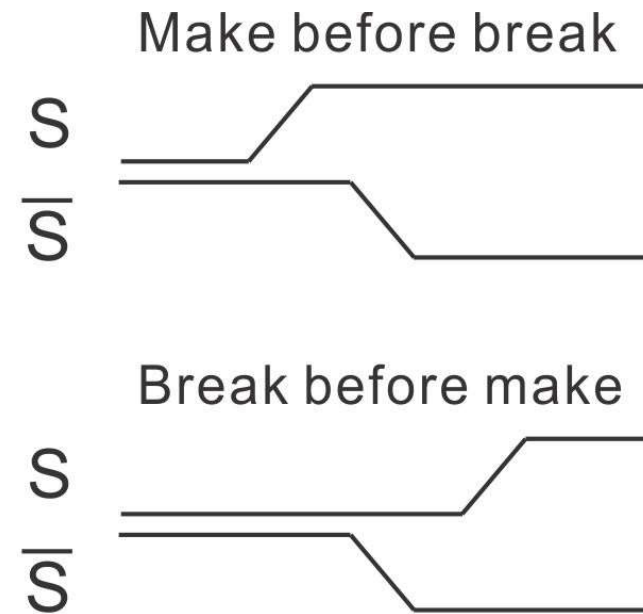
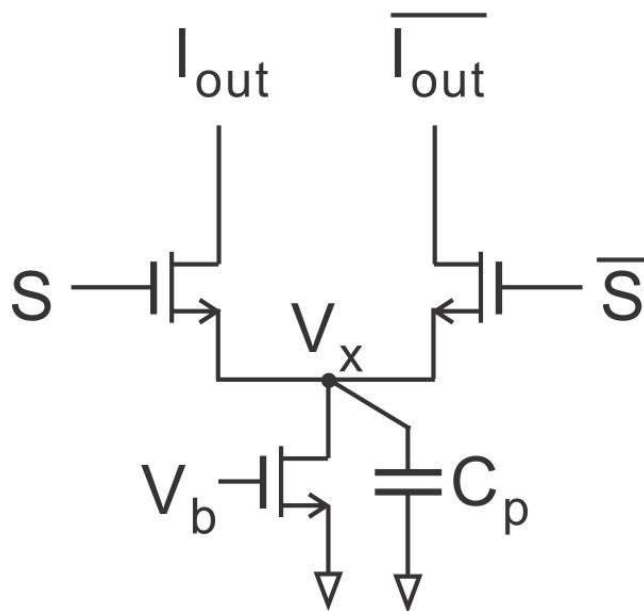
- Data retiming
 - Use a global clock to resample data near the current cells, so that they switch simultaneously independent of encoder delays
- Make before break
 - Ensure that the current is never turned off
 - S and S_bar are not simultaneously off
- Low swing driver
 - Drive differential pair with low swing to minimize coupling from control signals to output

Data Retiming



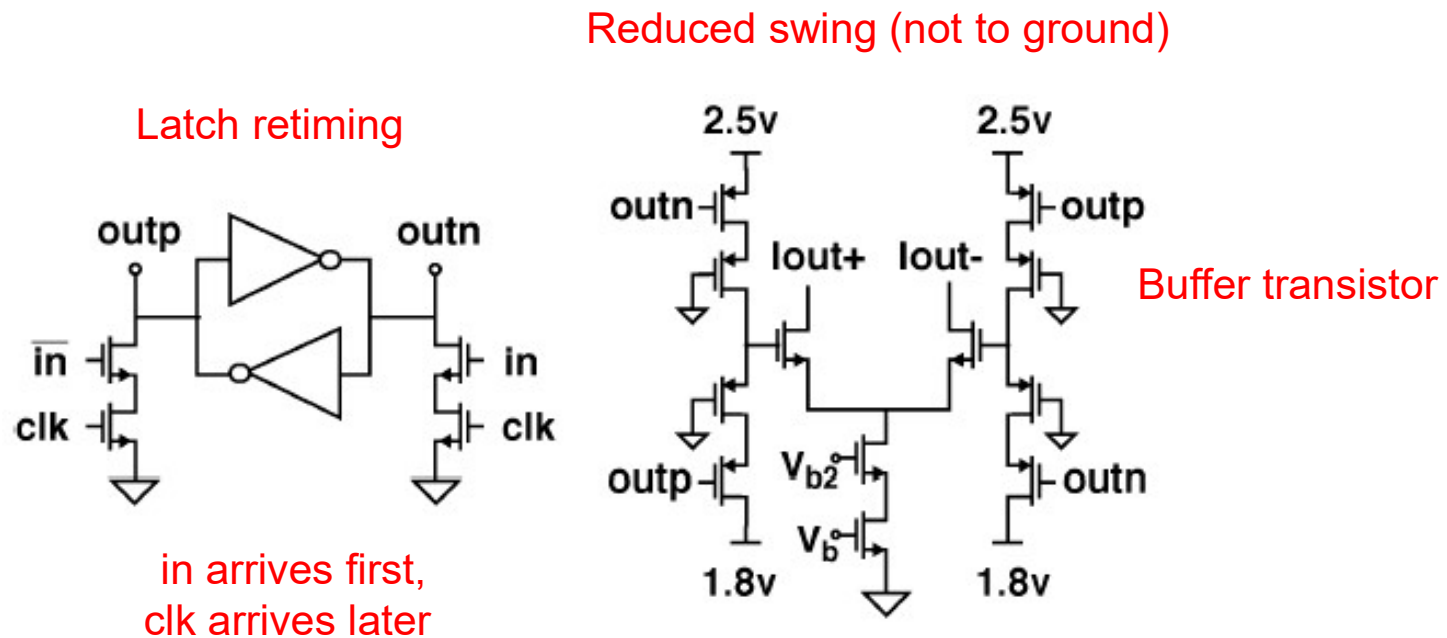
- Use a segmented DAC will help too.

Make Before Break



- Ensure that the current source is never turned off

Example Current Cell Implementation



[Barkin & Wooley, JSSC 4/2004]

Overview

- Topologies
- Current steering DAC architectures
- Static errors
- Dynamic errors
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High Performance DAC Example #1

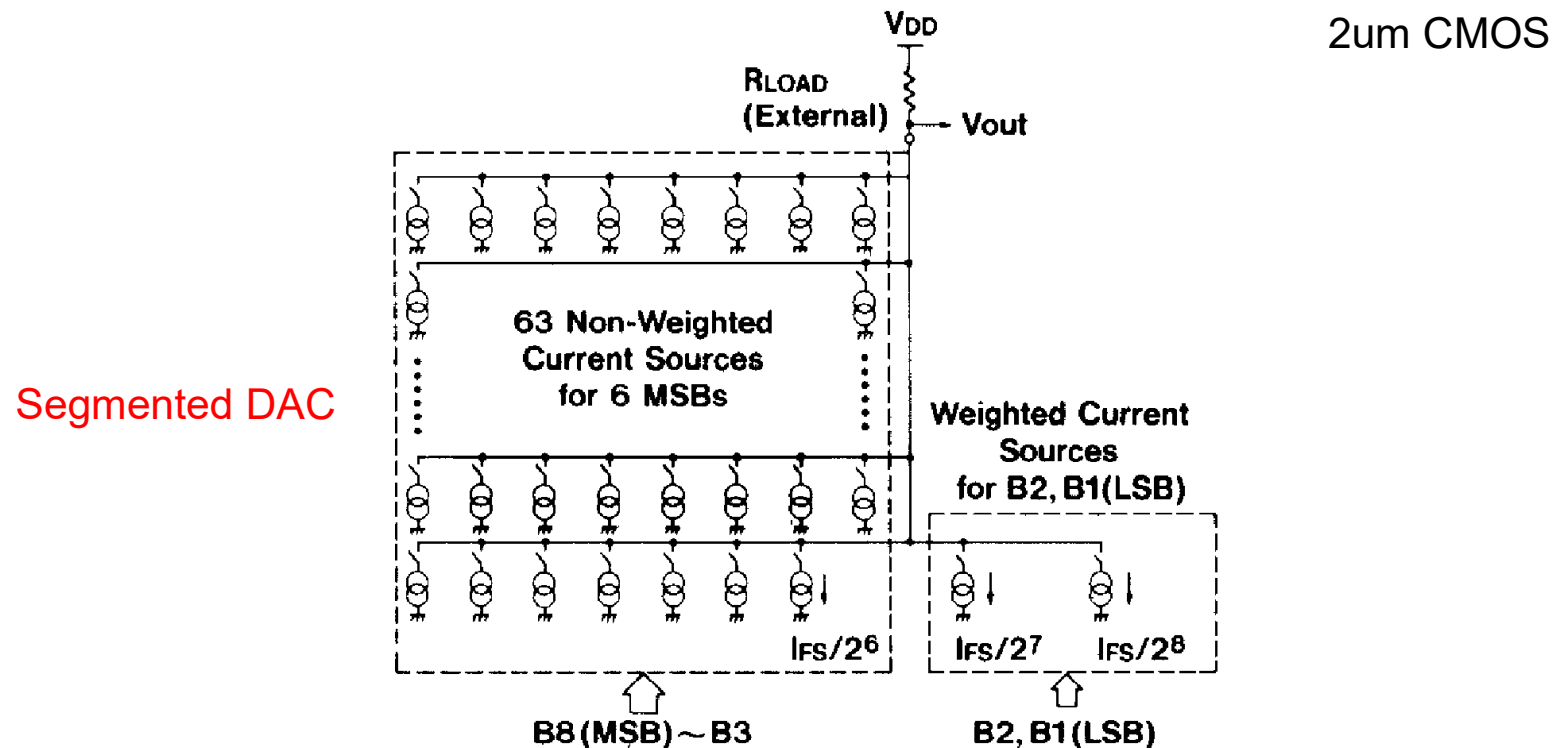


Fig. 1. Basic architecture of the DAC.

[T. Miki, Y. Nakamura, M. Nakaya, S. Asai, Y. Akasaka, and Y. Horiba, "An 80-MHz 8-bit CMOS D/A Converter," IEEE J. of Solid-State Circuits, pp. 983-988, Dec. 1986.]

Decoder Design

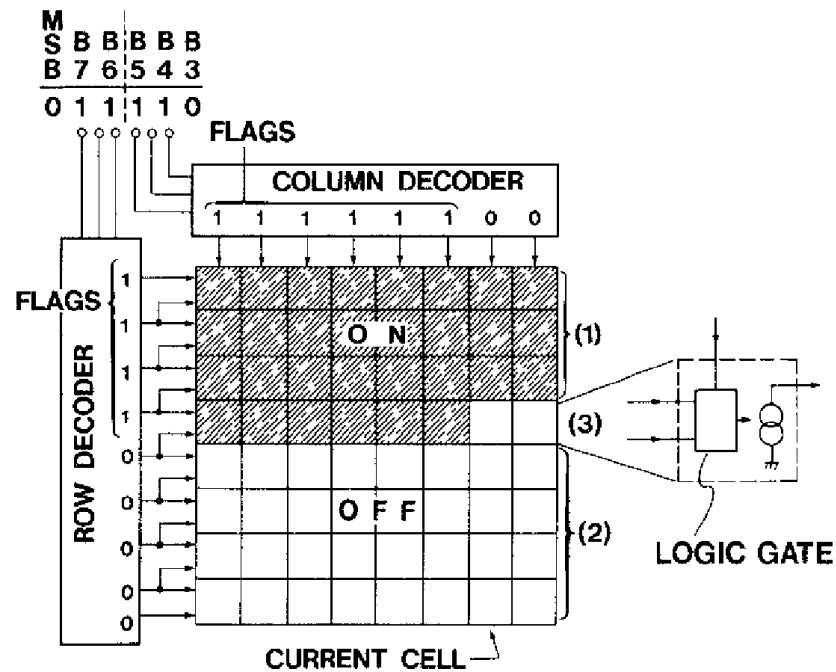


Fig. 2. Two-step decoding.

Fast decoding

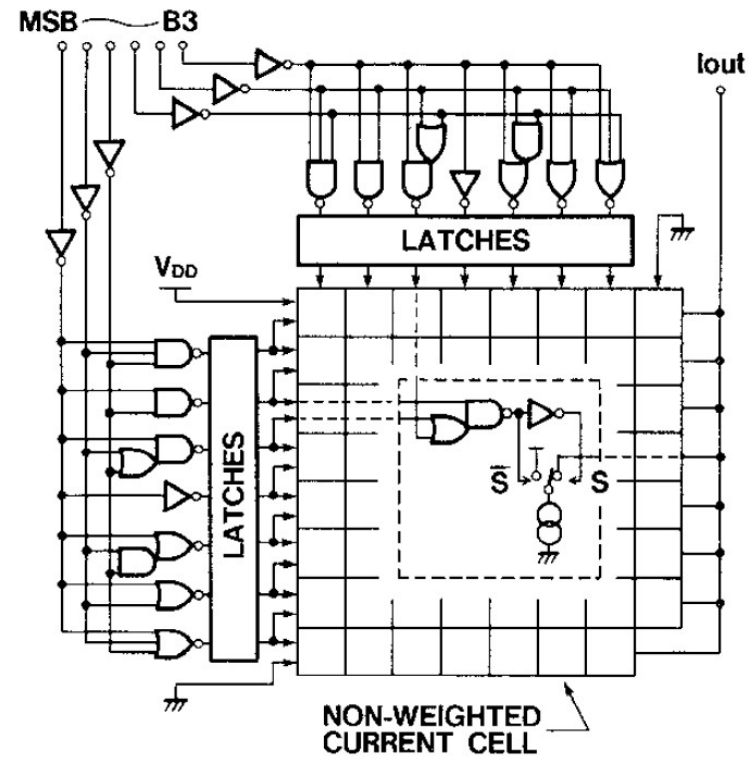
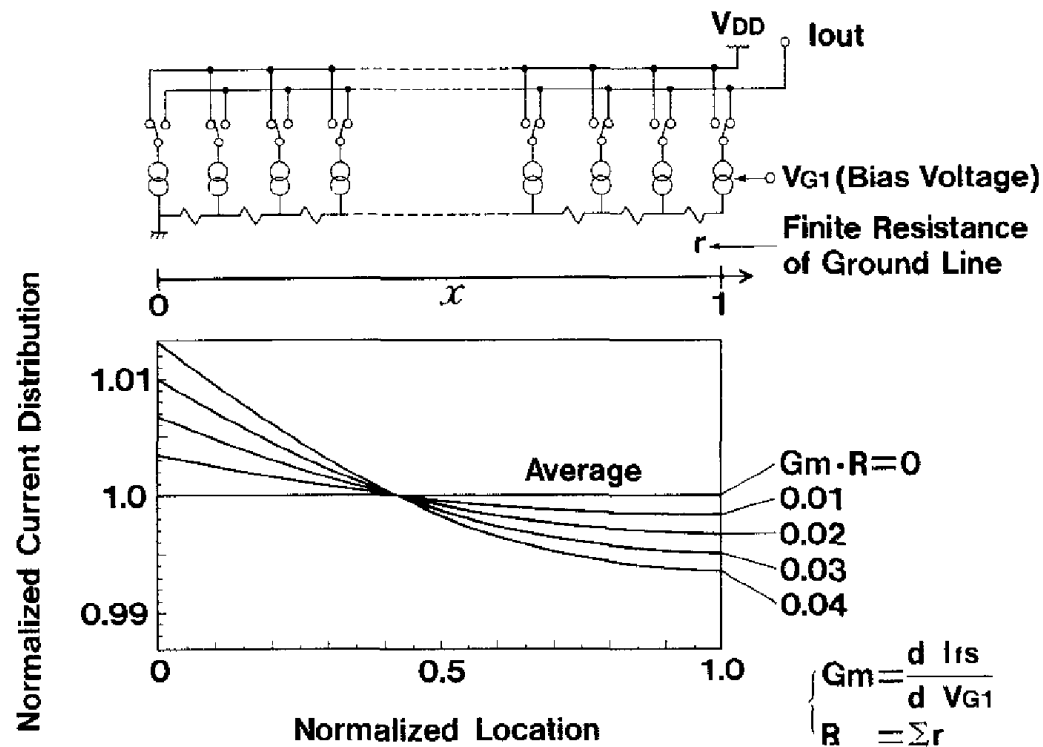


Fig. 3. High-speed decoding circuit.

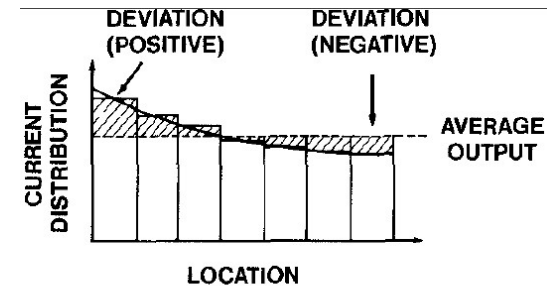
Latch retiming

Mitigating IR Drop

Systematically decreasing currents

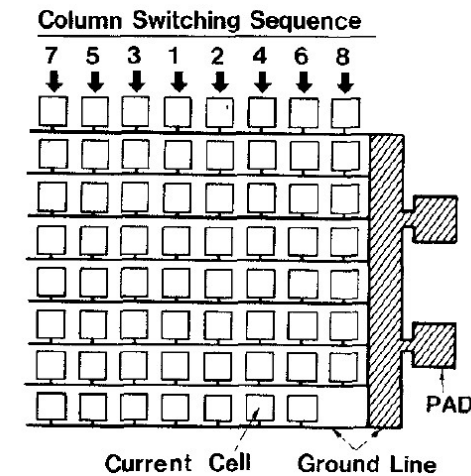


INL can be reduced by re-arranging DNLs

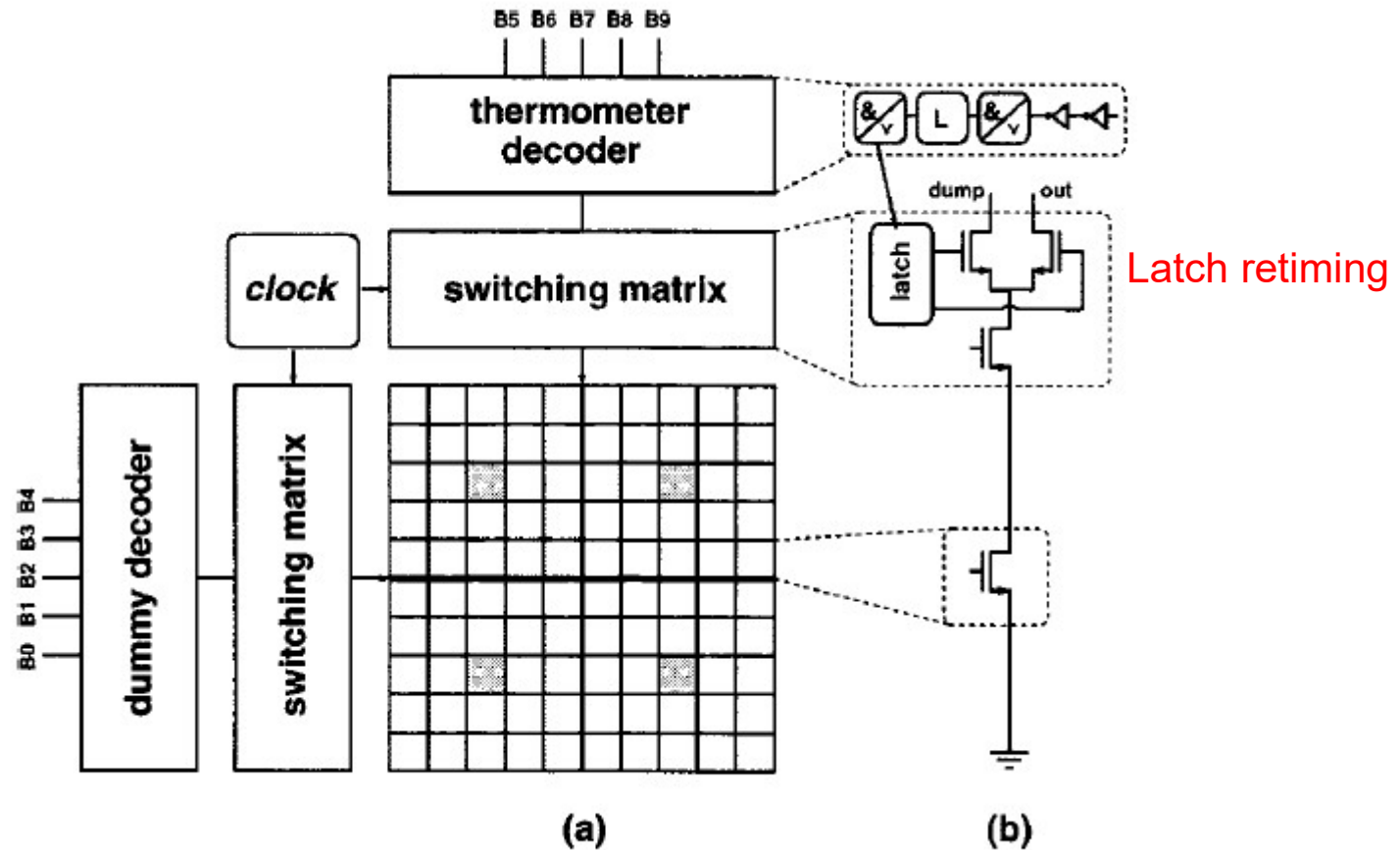


Reducing INL

Fig. 9. Symmetrical switching.



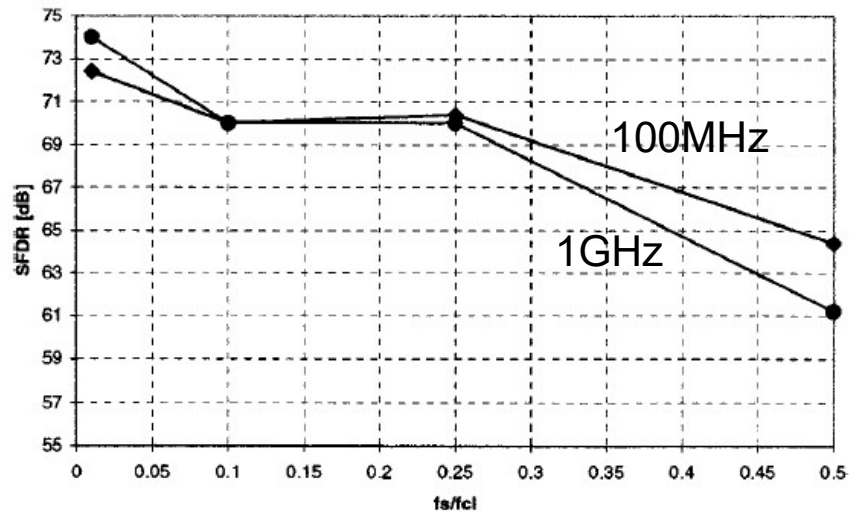
High Performance DAC Example #2



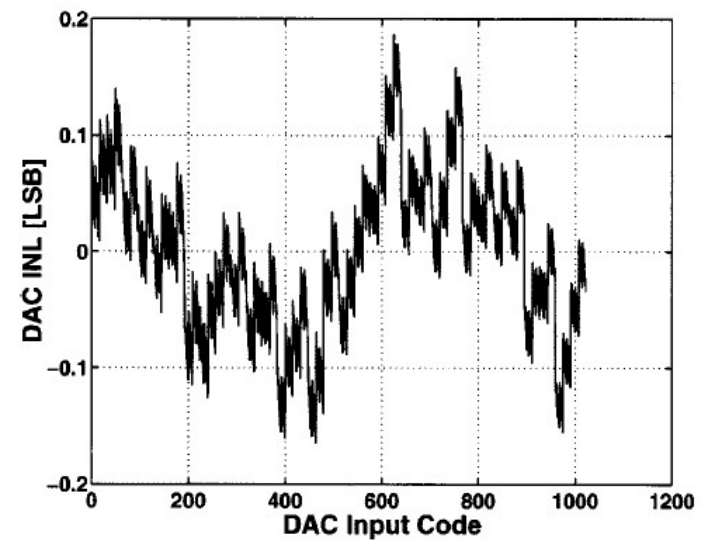
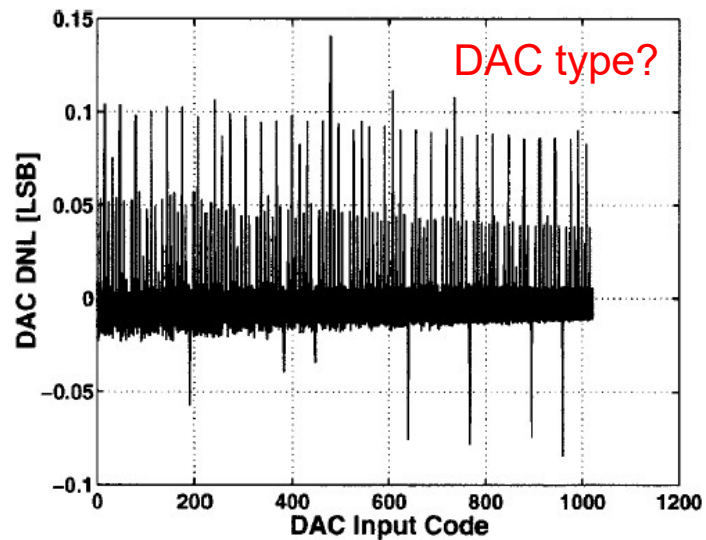
[Van den Bosch, JSSC 3/2001]

[Van den Bosch, JSSC 3/2001]

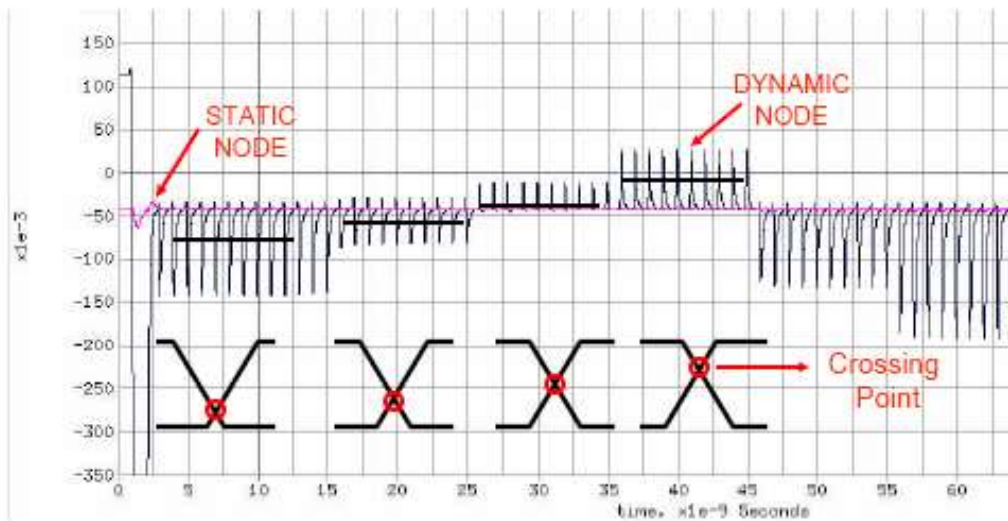
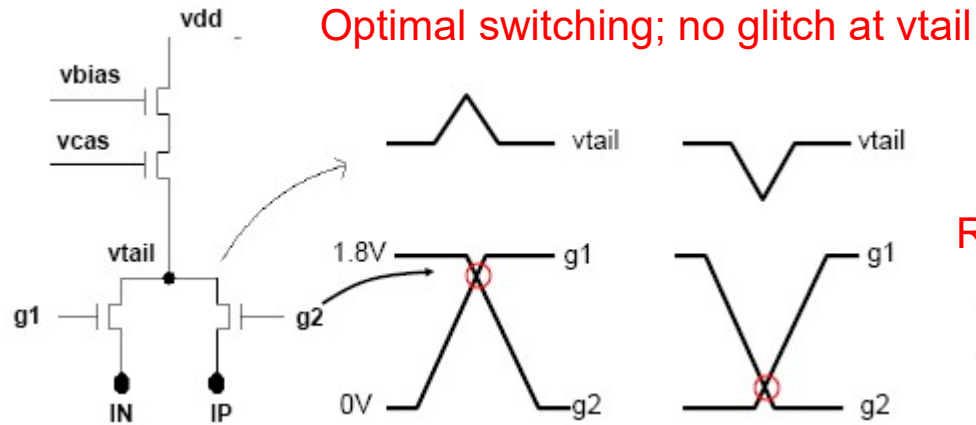
Measurements



Resolution	10 bit
Update rate	Up to 1 GS/s
INL	< 0.2LSB
DNL	<0.15 LSB
SFDR (490MHz@1GS/s)	61.2 dB
Analog/digital voltage supply	3V / 1.9 V
Power consumption	110mW (490Mhz@1GHz)
Active area	0.35 mm ²
Process	0.35 μm CMOS

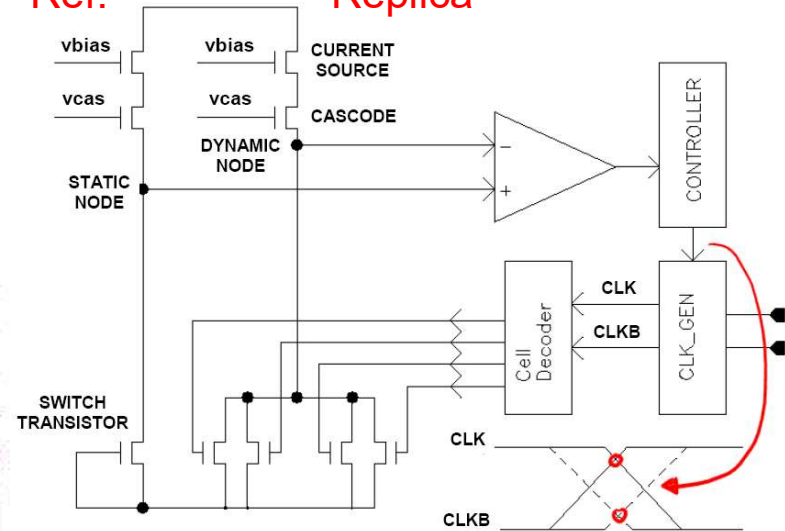


High Performance DAC Example #3



Gate Drive Crossover Control Loop

Ref. Replica

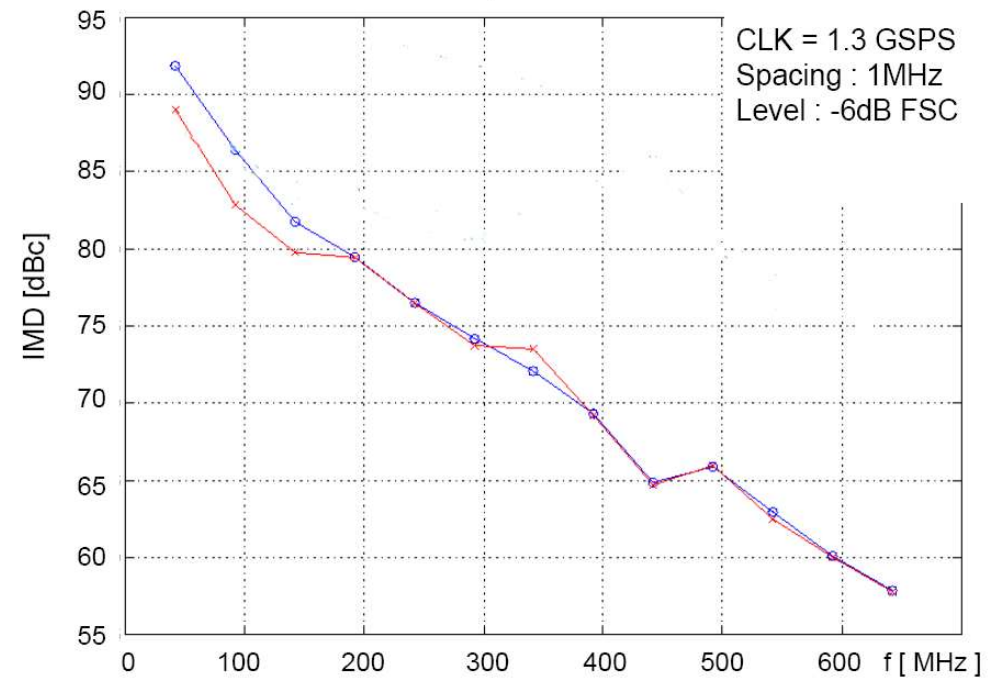


[Schaffner, ISSCC 2004]

Measurements

Max Sample Frequency	1.4	GSPS
Resolution	14	Bit
DNL	+/- 0.8	LSB
INL	+/- 2.1	LSB
SFDR @ 1.0 GSPS	> 60	dB
IMD @ 1.0 GSPS	> 64	dBc
NSD @ $f_{out} = 400\text{MHz}$	-155	dBm/Hz
Power (Core) @1.4GSPS	200	mW
Power(Total) @ 1.4GSPS	400	mW
Area (Core)	0.8	mm ²
Area (Chip)	6.25	mm ²

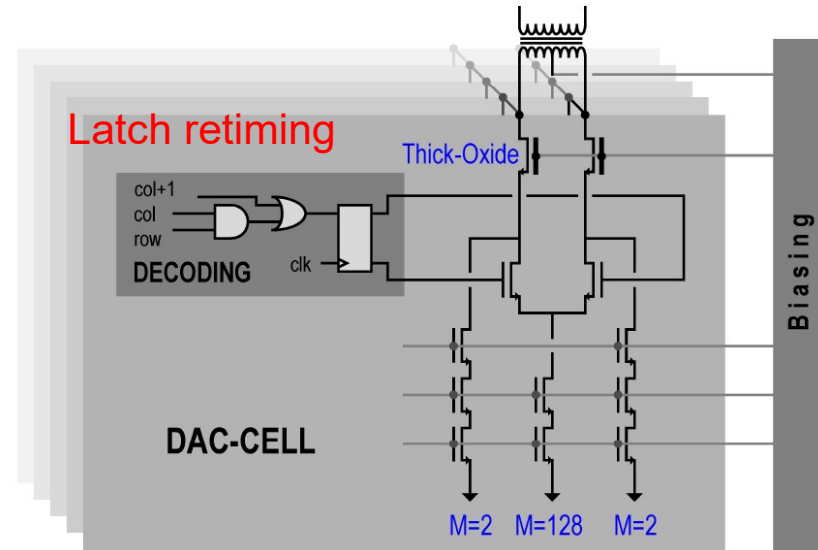
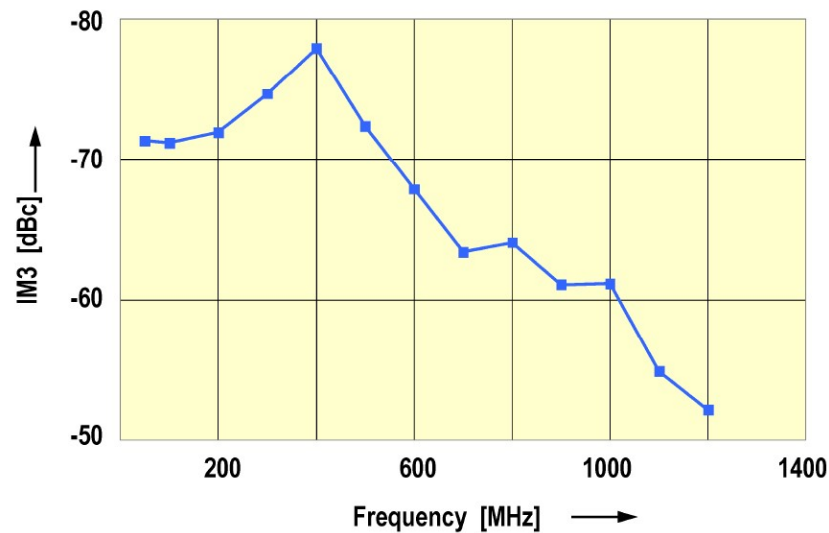
2 Tone Intermodulation Distortion



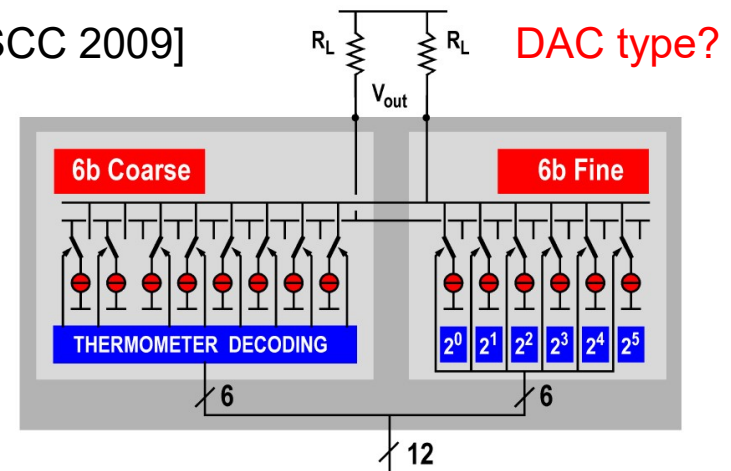
[Schafferer, ISSCC 2004]

High Performance DAC Example #4

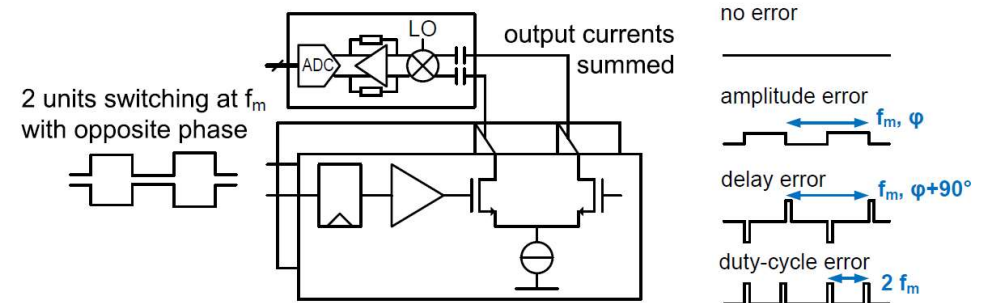
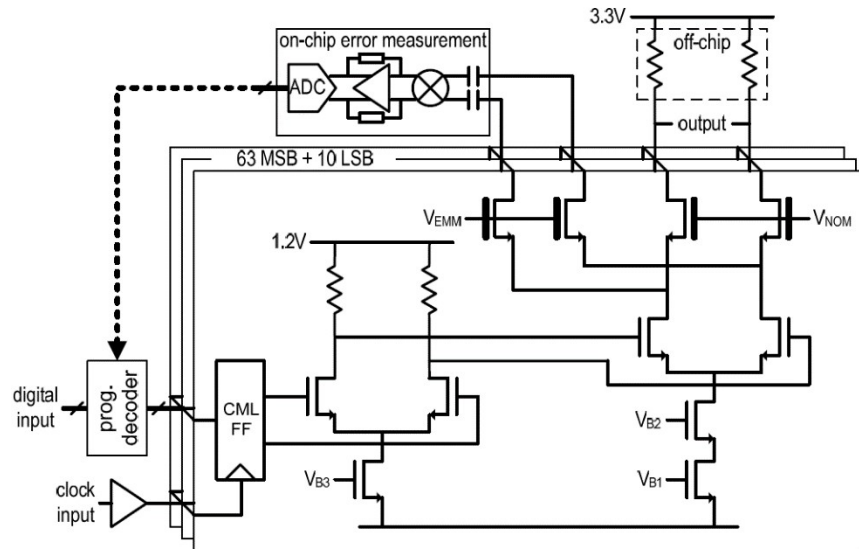
Technology	65nm CMOS
Number of Bits	12
Conversion speed	2.9 GS/s
INL	0.5 lsb
DNL	0.3 lsb
R_{load}	50 Ω
$I_{tail,total}$	50 mA
Voltage Swing	2.5 V_{ppd}
Power Dissipation	188 mW
Area	0.31 mm ²



[Lin, ISSCC 2009]



High Performance DAC Example #5



Error signal down-converted to DC, filtered and digitized

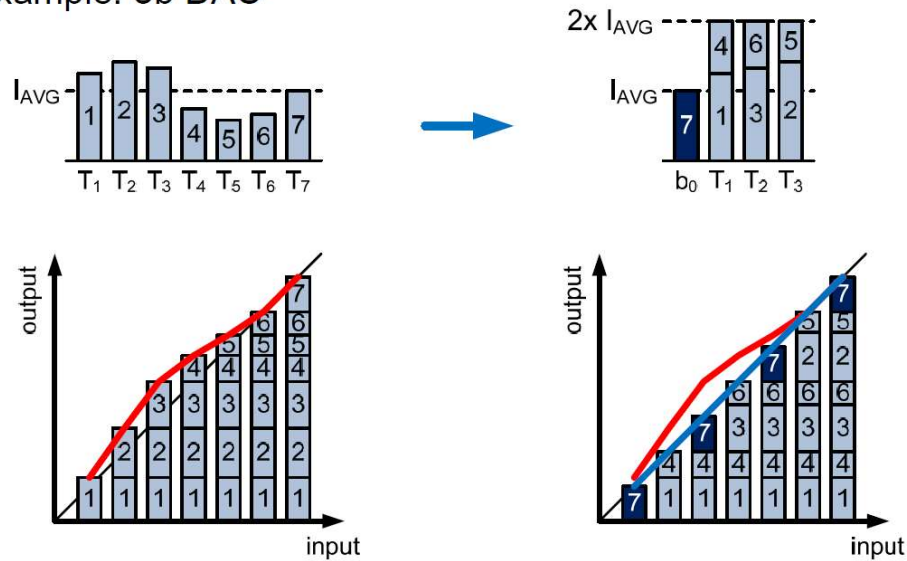
- $f_{LO}=f_m$ & $\phi_{LO}=\phi \Rightarrow$ amplitude error
- $f_{LO}=f_m$ & $\phi_{LO}=\phi+90^\circ \Rightarrow$ delay error
- $f_{LO}=2 f_m \Rightarrow$ duty-cycle error

[van de Vel, ISSCC 2014]

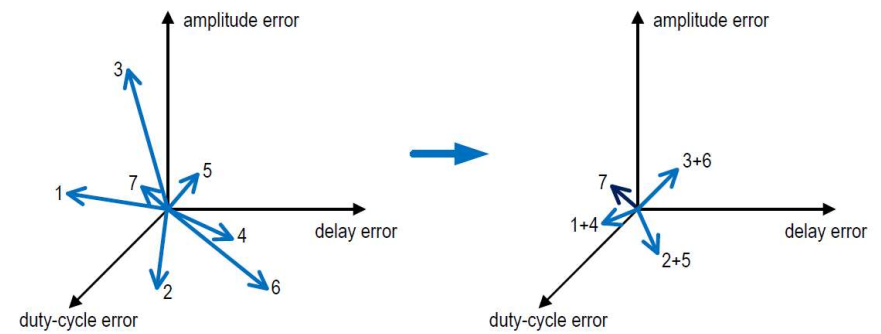
[Tang, JSSC11]

Sort and combine

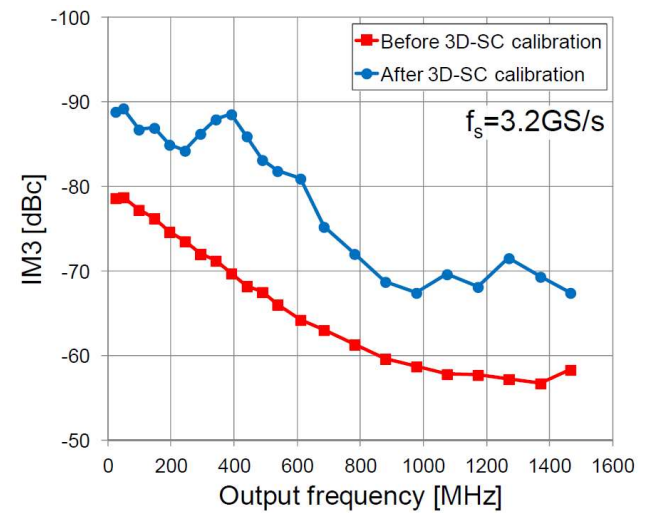
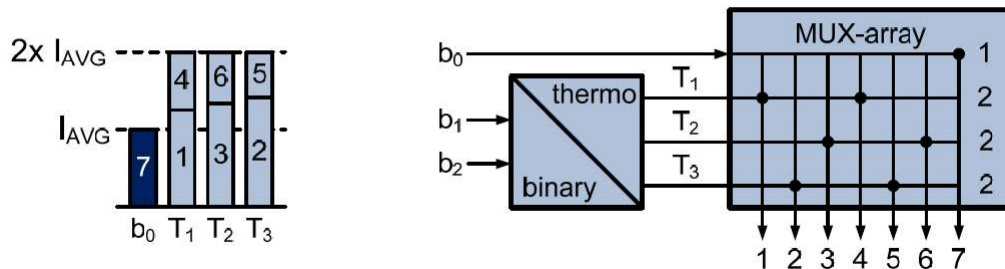
Example: 3b DAC



3D unit matching errors represented as 3D error vectors:

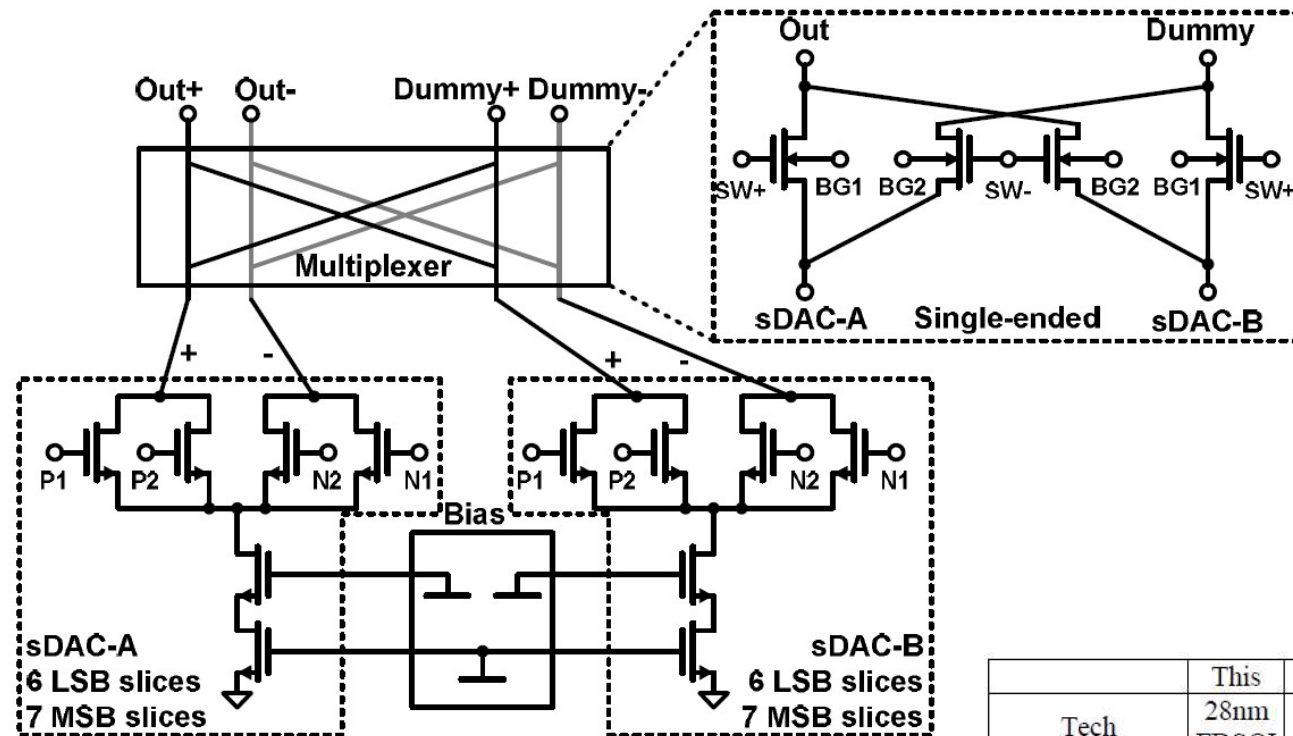


Sort-and-combine to combinations of 2 units:



[van de Vel, ISSCC 2014]

High Performance DAC Example #6



[Nauta, VLSI 2014]

	This	[1]	[2]	[3]	[4]	[5]
Tech	28nm FDSOI	28nm	180nm	65nm	65nm	65nm
Resolution [Bits]	9	13	14	12	9	6
Power [mW]	110	375	>600	70	60	750
V_{supply} [V]	1.0	1.8	-1.5/1.8	1.2	1.2	1.1/2.5
Area [mm^2]	0.04	1.16	4	0.4	0.04	0.24
Swing [$V_{\text{op-diff}}$]	0.425	N/A ^b	1.0	0.5	0.4	0.6
F_s [GHz]	11	9	6	1.7	3	56
SFDR ^a [dB]	51	-	52	58	49	43
IM3 ^a [dBc]	-51	-44	-65	-62.5	-60	-